

Derived Formulas and Corollaries in the Void Dynamics Model

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1 CF000 - Primitive Origin

1. Overall derived stage closure

$$0D \rightarrow 1D \rightarrow 2D.$$

What it is: The declared route that CF000 proves and closes.

2. Differentiated articulation

$$\text{Diff}(x) := \text{Real}(x) \wedge \neg \text{Flat}(x).$$

What it is: Derived definition of differentiated articulation.

3. Lowest-resolution invariant-bearing differentiated degree

$$\text{Deg}_{\min}(x) := \text{Inv}(x) \wedge \text{Diff}(x).$$

What it is: Derived definition of the first invariant-bearing differentiated degree.

4. No third terminal primitive pole survives

$$t = 0 \quad \text{or} \quad t = 1, \quad \text{and not both.}$$

What it is: Formula form of Theorem 3.5.2 for any terminal primitive pole-candidate t .

5. Nullity exclusion, applied to the realized witness

$$\omega \neq 0.$$

What it is: Derived consequence used in the forced-origin proof: the realized branch-origin cannot be absolute nullity.

6. Undifferentiated-totality exclusion, applied to the realized witness

$$\omega \neq 1.$$

What it is: Derived consequence used in the forced-origin proof: the realized branch-origin cannot be absolute undifferentiated totality.

7. Forced admissible origin

$$\text{Inv}(\omega).$$

What it is: Theorem 3.6.1. The realized origin is one primitive condition bearing unresolved pole-opposition internally.

8. Invariant-bearing precludes flatness

$$\forall x \in \mathcal{C}, \quad \text{Inv}(x) \Rightarrow \neg \text{Flat}(x).$$

What it is: Theorem 3.7.1. Any invariant-bearing condition is non-flat.

9. **Breakdown of perfect flatness at the realized origin**

$$\neg \text{Flat}(\omega).$$

What it is: Theorem 3.8.1. The realized origin is non-flat.

10. **Differentiated articulation at the realized origin**

$$\text{Diff}(\omega).$$

What it is: Corollary 3.8.2. The realized origin is differentiated articulation.

11. **Conjunction used to witness the lowest-resolution invariant-bearing degree**

$$\text{Inv}(\omega) \wedge \text{Diff}(\omega).$$

What it is: The conjunction immediately used in the proof of Corollary 3.8.3.

12. **Existence of a lowest-resolution invariant-bearing differentiated witness**

$$\exists x \in \mathcal{C} (\text{Orig}(x) \wedge \text{Deg}_{\min}(x)), \quad \text{with witness } x = \omega.$$

What it is: Corollary 3.8.3. The branch has a witnessed $\text{Deg}_{\min}$.

13. **Witness form at the realized origin**

$$\text{Deg}_{\min}(\omega).$$

What it is: Immediate witness-form consequence of Corollary 3.8.3.

14. **Constitutive invariant-bearing of admission**

if x is an admitted branch articulation, then $\text{Inv}(x)$.

What it is: Formula form equivalent to Theorem 4.3.1, which the paper states as: $\neg \text{Inv}(x)$ is contradictory.

15. **Contextual noncoincidence**

$$D_1 \neq D_2.$$

What it is: Theorem 4.5.1. Distinct higher-resolution determinations at the same level are non-identical.

16. **Closure of the first stage transition**

$$0D \rightarrow 1D.$$

What it is: Theorem 4.8.2. Forced origin plus non-flat differentiated articulation closes the move into first-axis articulation.

17. **Closure of the second stage transition**

$$1D \rightarrow 2D.$$

What it is: Theorem 4.8.3. Same-axis saturation plus orthogonal re-articulation closes the move into two-axis structure.

18. **Ordered-pair description at algebra unlock**

$$(a, b).$$

What it is: Theorem 4.9.1. Once 2D is admitted, two-axis placement licenses ordered-pair description.

19. **Worked section: ORS expressed algebraically — first axis-step**

$$R(1, 0) = (0, 1).$$

What it is: In Section 7.4.1, the rotation operator sends the first axis unit to the second axis unit.

20. **Worked section: ORS expressed algebraically — second axis-step**

$$R(0, 1) = (-1, 0).$$

What it is: In Section 7.4.1, the next orthogonal step lands on the inverted first-axis unit.

21. **Worked section: quarter-turn applied twice**

$$R^2(1, 0) = (-1, 0).$$

What it is: In Section 7.4.1, two axis-steps send the first-axis unit to its opposite.

22. **Worked section: operator-square relation**

$$R^2 = -1 \quad \text{on the first-axis unit.}$$

What it is: Section 7.4.1 identifies the rotation operator by the defining square relation of the imaginary unit.

23. **Worked section: imaginary-unit identification**

$$i^2 = -1.$$

What it is: Section 7.4.1. The paper identifies the ORS rotation operator with the imaginary unit.

24. **Worked section: explicit identification of the operator**

$$R = i.$$

What it is: Section 7.4.1. The orthogonal-rotation operator is recognized as i .

25. **Worked section: factorial identity**

$$0! = 1.$$

What it is: Section 7.4.2. The empty arrangement is the unique identity arrangement.

26. **Empirical consistency calculation: logarithm of carrier size**

$$\ln(5000) = 8.517.$$

What it is: Section 6.2 numerical consistency calculation.

27. **Empirical consistency calculation: observed depth ratio**

$$\frac{N_{\max}}{\ln L} = \frac{9}{\ln(5000)} \approx \frac{9}{8.517} \approx 1.0567.$$

What it is: Readable reconstruction of the manuscript's line reporting the depth ratio for $L = 5000$ and $N_{\max} = 9$.

28. **Worked section, explicitly marked out of scope**

$$a^2 + b^2 = c^2.$$

What it is: Section 7.4.3 mentions the Pythagorean theorem only as a boundary marker. The paper explicitly says this is *not* derived in CF000.

2 CF00 - Induced Geometry and Emergent Dynamics

1. **Real completion forced from the CF000 handoff**

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}.$$

What it is: Theorem 2.5.1. The completion chain forced from proto- $\mathbb{N}$.

2. **Complex field inherited from 2D ORS algebra**

$$\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}.$$

What it is: Theorem 2.5.2. The natural two-axis algebra licensed by CF000.

3. **Imaginary-unit relation carried forward from CF000**

$$i^2 = -1.$$

What it is: Theorem 2.5.2 restates the ORS algebraic relation inherited from CF000.

4. **Polynomial presentation of the complex field**

$$\mathbb{C} = \mathbb{R}[i]/(i^2 + 1).$$

What it is: Theorem 2.5.2. Equivalent algebraic presentation of $\mathbb{C}$.

5. **Completion metric on the real line**

$$d(a, b) = |a - b|.$$

What it is: Definition/Theorem 2.5.3. The metric generating the natural topology on $\mathbb{R}$.

6. **Continuous completion of ORS**

$$R_0 = I, \quad R_{\theta+\phi} = R_\theta R_\phi.$$

What it is: Theorem 2.5.3b.1. The continuous one-parameter completion of the discrete ORS action.

7. **Discrete ORS seed carried into the continuous completion**

$$R^2 = -I.$$

What it is: Theorem 2.5.3b.1. The opposite-pole square relation inherited from CF000.

8. **Primitive half-turn constant**

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

What it is: Definition 2.5.3b.2. The first positive rotation parameter reaching the opposite pole.

9. **Opposite-pole hit set**

$$H := \{\theta > 0 : R_\theta = -I\}.$$

What it is: Theorem 2.5.3b.3. The set used to prove existence and uniqueness of the least positive half-turn.

10. **Opposite-pole hit set bounded away from zero**

$$H \subseteq [\varepsilon, \infty).$$

What it is: Theorem 2.5.3b.3. The proof step establishing a positive lower bound for the hit set.

11. **Geometric realization of the half-turn constant**

$$\pi = \iint_{D^2} dx dy, \quad D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

What it is: Corollary 2.5.3b.4. The same π realized as the measure of the unit disk.

12. **Carrier domain**

$$\mathcal{M}.$$

What it is: Theorem 2.5.5. The first admissible connected C^1 host of continued articulation past 2D.

13. **Effective Hilbert bundle over the carrier**

$$\pi : \mathcal{H} \rightarrow \mathcal{M}.$$

What it is: Theorem 2.5.6. The minimal complex Hilbert bundle forced by the representative ontology.

14. **Local $U(1)$ redundancy**

$$|\psi(x)\rangle \sim e^{i\Lambda(x)}|\psi(x)\rangle, \quad \Lambda \in C^1(\mathcal{M}, \mathbb{R}).$$

What it is: Theorem 2.5.7. Local phase redundancy of normalized complex fibers.

15. **Carrier-domain distance notation**

$$d_{\mathcal{M}}.$$

What it is: Section 2.6. The derived carrier-domain distance induced from the real completion.

16. **Projector onto the representative line**

$$P_{\parallel} := |\psi\rangle\langle\psi|.$$

What it is: Section 4.1. The vertical projector.

17. **Complementary horizontal projector**

$$P_{\perp} := 1 - |\psi\rangle\langle\psi|.$$

What it is: Section 4.1. The projector removing self-parallel representative content.

18. **Variation decomposition**

$$v = v_{\parallel} + v_{\perp}, \quad v_{\parallel} = P_{\parallel}v, \quad v_{\perp} = P_{\perp}v.$$

What it is: Section 4.1. Decomposition of local variation into vertical and horizontal parts.

19. **Local rephasing**

$$|\psi\rangle \mapsto e^{i\Lambda}|\psi\rangle.$$

What it is: Section 4.2. The local gauge redundancy acting on representatives.

20. **Projected local derivative**

$$|\partial_\mu \psi\rangle_\perp := P_\perp |\partial_\mu \psi\rangle = |\partial_\mu \psi\rangle - |\psi\rangle\langle\psi|\partial_\mu \psi\rangle.$$

What it is: Section 5.1. The quotient-physical local derivative.

21. **Quantum geometric tensor**

$$Q_{\mu\nu} = \langle\partial_\mu \psi | \partial_\nu \psi\rangle - \langle\partial_\mu \psi | \psi\rangle\langle\psi | \partial_\nu \psi\rangle.$$

What it is: Section 5.2 and Theorem 5.4.1. The first induced gauge-invariant bilinear geometric object.

22. **Equivalent projected form of the QGT**

$$Q_{\mu\nu} = \langle\partial_\mu \psi | P_\perp \partial_\nu \psi\rangle.$$

What it is: Section 5.2. The QGT written directly in terms of projected derivatives.

23. **Gauge transformation of the raw derivative**

$$|\partial_\mu \psi\rangle \mapsto e^{i\Lambda} (|\partial_\mu \psi\rangle + i(\partial_\mu \Lambda)|\psi\rangle).$$

What it is: Section 5.3. The raw derivative picks up a vertical term under rephasing.

24. **Metric and curvature split from the QGT**

$$g_{\mu\nu} := \text{Re } Q_{\mu\nu}, \quad \Omega_{\mu\nu} := -2 \text{Im } Q_{\mu\nu}.$$

What it is: Section 5.5 and Theorem 5.6.1. The induced metric and curvature sectors.

25. **Recombination of the QGT from its two sectors**

$$Q_{\mu\nu} = g_{\mu\nu} - \frac{i}{2}\Omega_{\mu\nu}.$$

What it is: Section 5.5. The Hermitian tensor decomposed into symmetric and antisymmetric pieces.

26. **Normalization constraint**

$$\langle\psi, \psi\rangle = 1.$$

What it is: Section 6.1. The normalized representative condition.

27. **Time derivative of normalization**

$$\frac{d}{dt}\langle\psi, \psi\rangle = \langle\dot{\psi}, \psi\rangle + \langle\psi, \dot{\psi}\rangle = 2 \text{Re}\langle\psi, \dot{\psi}\rangle = 0.$$

What it is: Section 6.1. The step showing the representative tangent has purely imaginary self-overlap.

28. **Vertical phase scalar**

$$\langle \psi, \dot{\psi} \rangle = -i \alpha[\psi].$$

What it is: Section 6.1. Definition of the real scalar functional controlling the vertical phase motion.

29. **Horizontal tangent component**

$$\chi[\psi] := \dot{\psi} + i \alpha[\psi] \psi.$$

What it is: Section 6.1. The quotient-physical component of representative evolution.

30. **Orthogonality of the horizontal tangent**

$$\langle \psi, \chi[\psi] \rangle = \langle \psi, \dot{\psi} \rangle + i \alpha[\psi] \langle \psi, \psi \rangle = -i \alpha[\psi] + i \alpha[\psi] = 0.$$

What it is: Section 6.1. The horizontal component is orthogonal to the representative line.

31. **Unique tangent decomposition of norm-preserving representative evolution**

$$\dot{\psi} = -i \alpha[\psi] \psi + \chi[\psi], \quad \langle \psi, \chi[\psi] \rangle = 0.$$

What it is: Section 6.1. The vertical-plus-horizontal split of representative dynamics.

32. **Bundle maps induced by curvature and metric**

$$J := \Omega^\sharp, \quad M := g^\sharp.$$

What it is: Section 6.3. The reversible and irreversible operators induced from quotient geometry.

33. **Mixed scalar-generated horizontal law**

$$\bar{\chi} = J d\mathcal{I} + M d\Sigma = \Omega^\sharp d\mathcal{I} + g^\sharp d\Sigma.$$

What it is: Theorem 6.5.1. The emergent scalar-generated mixed dynamical architecture.

34. **Minimal conservative/entropy-bearing constitutive generator**

$$\mathcal{I}[\psi, u] = \int_{\mathcal{M}} \left(\frac{\tau}{2} |D_t \psi_\perp|_g^2 + \frac{D}{2} \sum_{i=1}^d |D_i \psi_\perp|_g^2 + V([\psi]) + u \right) d\mu.$$

What it is: Section 7.2. The narrowest admissible invariant generator at this stage.

35. **Entropy generator**

$$\Sigma[\psi, u] = \int_{\mathcal{M}} s(u) d\mu.$$

What it is: Section 7.2. The entropy-bearing scalar generator.

36. **Entropy-chart regularity and monotonicity conditions**

$$s \in C^2, \quad s'(u) > 0, \quad s''(u) < 0.$$

What it is: Section 7.2 and Theorem 7.5.1. The theorem-grade constitutive equivalence class.

37. **Conservative-plus-reservoir split used in the conjugacy lemma**

$$\mathcal{I}[\psi, u] = \mathcal{I}_0[\psi] + \int_{\mathcal{M}} u d\mu, \quad \Sigma[\psi, u] = \int_{\mathcal{M}} s(u) d\mu.$$

What it is: Section 7.4. Starting point for the reservoir-coordinate conjugacy theorem.

38. **Reservoir-coordinate transformation**

$$\Phi : (\psi, u) \mapsto (\psi, \sigma), \quad \sigma = s(u).$$

What it is: Lemma 7.4.1. Smooth monotone change of entropy chart.

39. **Transformed generators under reservoir-coordinate change**

$$\tilde{\mathcal{I}}[\psi, \sigma] = \mathcal{I}_0[\psi] + \int_{\mathcal{M}} s^{-1}(\sigma) d\mu, \quad \tilde{\Sigma}[\psi, \sigma] = \int_{\mathcal{M}} \sigma d\mu.$$

What it is: Lemma 7.4.1. The conjugate constitutive pair in the new chart.

40. **Conjugate vector field**

$$\tilde{X} = T\Phi X \Phi^{-1}.$$

What it is: Lemma 7.4.1. The transformed law under the reservoir-coordinate diffeomorphism.

41. **Transformed entropy along corresponding trajectories**

$$\tilde{\Sigma}[\psi, \sigma] = \Sigma[\psi, s^{-1}(\sigma)].$$

What it is: Lemma 7.4.1. Entropy functional in the transformed chart.

42. **Entropy-production invariance under conjugacy**

$$\frac{d}{dt} \tilde{\Sigma} = \frac{d}{dt} \Sigma.$$

What it is: Lemma 7.4.1. The sign of entropy production is preserved under smooth monotone reservoir reparameterization.

43. **Conjugate solution operator**

$$\tilde{U}_{t,t_0} = \Phi \circ U_{t,t_0} \circ \Phi^{-1}.$$

What it is: Lemma 7.4.1. The transformed nonlinear flow.

44. **Conjugate linearized propagator**

$$D\tilde{U}_{t,t_0} = D\Phi \circ DU_{t,t_0} \circ D\Phi^{-1}.$$

What it is: Lemma 7.4.1. The Fréchet derivative transforms covariantly under the chart change.

45. **Theorem-grade constitutive equivalence class**

$$s \in C^2, \quad s'(u) > 0, \quad s''(u) < 0.$$

What it is: Theorem 7.5.1. The constitutive content lives at the level of this equivalence class, not one special reservoir chart.

46. **Canonical execution specialization**

$$s(u) = k_B \log u, \quad u > 0.$$

What it is: Section 7.6. A canonical specialization inside the theorem-grade equivalence class.

47. **Nonlinear solution operator on primitive state space**

$$U_{t,t_0} : Z \rightarrow Z.$$

What it is: Section 8.1. The closed primitive-law flow on admissible quotient states.

48. **Linearized retarded propagator**

$$DU_{t,t_0}|_{z(t_0)}.$$

What it is: Section 8.1. The Fréchet derivative of the nonlinear solution operator at the initial state.

49. **Derived support relation**

$$(x_0, t_0) \rightsquigarrow (x, t)$$

if and only if there exists a localized horizontal perturbation δz_0 near x_0 such that

$$(DU_{t,t_0}|_{z(t_0)} \delta z_0)(x) \neq 0.$$

What it is: Section 8.2 and Theorem 8.3.1. The locality-bearing support relation derived from propagation under the law.

50. **Exact support set**

$$\text{Supp}(t; t_0, x_0) = \overline{\{x \in \mathcal{M} : (x_0, t_0) \rightsquigarrow (x, t)\}}.$$

What it is: Section 8.2. The closure of the propagated-support region.

51. **Carrier-domain cone bound on support**

$$\text{Supp}(t; t_0, x_0) \subseteq \{x \in \mathcal{M} : d_{\mathcal{M}}(x, x_0) \leq c_{\text{eff}}(t - t_0)\}.$$

What it is: Theorem 8.3.1. Finite-speed support bound for telegraph-type and Cattaneo-type principal part.

52. **Local overlap map**

$$y \mapsto \langle \psi(x), \psi(y) \rangle.$$

What it is: Section 8.6. The overlap function used to derive local gauge-hosting neighborhoods.

53. **Nonvanishing-overlap neighborhood condition**

$$\langle \psi(x), \psi(y) \rangle \neq 0 \quad \text{for all } y \in U_x.$$

What it is: Lemma 8.6.1. Local open neighborhoods where overlap phases are well-defined.

54. **Normalized overlap phase**

$$\mathcal{U}(x, y) = \frac{\langle \psi(x), \psi(y) \rangle}{|\langle \psi(x), \psi(y) \rangle|}.$$

What it is: Lemma 8.6.1. The local link phase on overlap neighborhoods.

3 CF01 - QGT to Metriplectic Brackets

1. **Quantum geometric tensor**

$$Q_{\mu\nu}(R) = \langle \partial_\mu \psi \mid \partial_\nu \psi \rangle - \langle \partial_\mu \psi \mid \psi \rangle \langle \psi \mid \partial_\nu \psi \rangle.$$

What it is: Canonical QGT definition in Section 1.1.

2. **Normalized state condition**

$$\langle \psi \mid \psi \rangle = 1.$$

What it is: The normalization condition used in the QGT definition.

3. **QGT split into metric and curvature sectors**

$$Q_{\mu\nu} = g_{\mu\nu} - \frac{i}{2} \Omega_{\mu\nu}.$$

What it is: Fundamental decomposition of the QGT in Section 1.2.

4. **Quantum metric from the QGT**

$$g_{\mu\nu} = \text{Re}(Q_{\mu\nu}) = \frac{1}{2}(Q_{\mu\nu} + Q_{\nu\mu}^*).$$

What it is: Symmetric real sector of the QGT.

5. **Berry curvature from the QGT**

$$\Omega_{\mu\nu} = -2 \text{Im}(Q_{\mu\nu}) = i(Q_{\mu\nu} - Q_{\nu\mu}^*).$$

What it is: Antisymmetric imaginary sector of the QGT.

6. **Berry connection**

$$A_\mu(R) = i\langle\psi(R) | \partial_\mu\psi(R)\rangle.$$

What it is: Gauge-potential form of the Berry connection.

7. **Berry curvature as field strength**

$$\Omega_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = i(\langle\partial_\mu\psi | \partial_\nu\psi\rangle - \langle\partial_\nu\psi | \partial_\mu\psi\rangle).$$

What it is: Berry curvature expressed from the Berry connection and state derivatives.

8. **Berry curvature as a two-form**

$$\omega = \frac{1}{2}\Omega_{\mu\nu} dR^\mu \wedge dR^\nu.$$

What it is: Lemma 2.1. The Berry curvature interpreted as a symplectic two-form.

9. **Bianchi identity / closure condition**

$$\partial_\lambda\Omega_{\mu\nu} + \partial_\mu\Omega_{\nu\lambda} + \partial_\nu\Omega_{\lambda\mu} = 0.$$

What it is: Lemma 2.1. The closure condition used to support the symplectic interpretation.

10. **Non-degeneracy condition for the symplectic region**

$$\det(\Omega) \neq 0.$$

What it is: The local non-degeneracy condition required where the inverse Berry curvature is used.

11. **J-bracket from Berry curvature**

$$\{f, g\}_J = \Omega^{\mu\nu}(R) \partial_\mu f \partial_\nu g.$$

What it is: Definition of the J-bracket in Section 2.3.

12. **Jacobi identity for the J-bracket**

$$\{\{f, g\}_J, h\}_J + \{\{h, f\}_J, g\}_J + \{\{g, h\}_J, f\}_J = 0.$$

What it is: Theorem 2.1. The J-bracket satisfies Jacobi on the non-degenerate region.

13. **Inverse-curvature differentiation identity**

$$\partial_\alpha \Omega_{\mu\nu} J^{\nu\lambda} + \Omega_{\mu\nu} \partial_\alpha J^{\nu\lambda} = 0.$$

What it is: Coordinate proof step used in Remark 2.1a, where $J^{\mu\nu} = (\Omega^{-1})^{\mu\nu}$.

14. **Jacobi condition in Schouten form**

$$J^{\alpha\mu} \partial_\alpha J^{\nu\lambda} + J^{\alpha\nu} \partial_\alpha J^{\lambda\mu} + J^{\alpha\lambda} \partial_\alpha J^{\mu\nu} = 0.$$

What it is: Coordinate form of the Jacobi condition $[J, J]_{\text{Schouten}} = 0$.

15. **Hamiltonian flow generated by Berry curvature**

$$\dot{R}^\mu = \{R^\mu, H\}_J = \Omega^{\mu\nu} \partial_\nu H.$$

What it is: Reversible conservative evolution generated by the J-bracket.

16. **J-Casimir condition**

$$\{\Sigma_J, H\}_J = 0.$$

What it is: Casimir condition for a functional preserved by the J-bracket.

17. **Quantum metric in excited-state form**

$$g_{\mu\nu}(R) = \text{Re} \langle \partial_\mu \psi | \partial_\nu \psi \rangle = \sum_n \frac{|\langle n | \partial_\mu H | \psi \rangle|^2}{(E_\psi - E_n)^2}.$$

What it is: Explicit form of the quantum metric given in Section 3.1.

18. **Metric line element**

$$ds^2 = g_{\mu\nu} dR^\mu dR^\nu.$$

What it is: The parameter-space distance induced by the quantum metric.

19. **Quantum Fisher information relation**

$$g_{\mu\nu} = \text{Re} \langle \partial_\mu \psi | \partial_\nu \psi \rangle = \frac{1}{4} \text{Tr}(\rho \{L_\mu, L_\nu\}).$$

What it is: Lemma 3.1. Identification of the quantum metric with the quantum Fisher information metric.

20. **Density operator for the pure state**

$$\rho = |\psi\rangle\langle\psi|.$$

What it is: The density operator used in the Fisher-information formula.

21. **M-bracket from the quantum metric**

$$(f, g)_M = g_{\mu\nu}(R) \partial_\mu f \partial_\nu g.$$

What it is: Definition of the M-bracket in Section 3.3.

22. **Symmetry of the M-bracket**

$$(f, g)_M = (g, f)_M.$$

What it is: Theorem 3.1. Symmetry inherited from the symmetry of $g_{\mu\nu}$.

23. **Positive semi-definiteness of the M-bracket**

$$(f, f)_M \geq 0.$$

What it is: Theorem 3.1. Positivity property of the metric bracket.

24. **Degeneracy condition for the M-bracket**

$$(I, \cdot)_M = 0.$$

What it is: Theorem 3.1. Casimir degeneracy condition for the metric bracket.

25. **Gradient flow generated by the quantum metric**

$$\dot{R}^\mu = (R^\mu, S)_M = g^{\mu\nu} \partial_\nu S.$$

What it is: Dissipative/gradient evolution generated by the M-bracket.

26. **Monotone entropy production under metric flow**

$$\frac{dS}{dt} = g^{\mu\nu} \partial_\mu S \partial_\nu S \geq 0.$$

What it is: Entropy monotonicity under the pure M-flow.

27. **Combined metriplectic evolution**

$$\dot{R}^\mu = \{R^\mu, H\}_J + (R^\mu, S)_M = \Omega^{\mu\nu} \partial_\nu H + g^{\mu\nu} \partial_\nu S.$$

What it is: Section 4.1. Full metriplectic evolution equation.

28. **J-degeneracy condition for entropy**

$$\{S, \cdot\}_J = 0.$$

What it is: Entropy is a Casimir of the conservative limb.

29. **M-degeneracy condition for energy**

$$(H, \cdot)_M = 0.$$

What it is: Energy is a Casimir of the dissipative limb.

30. **Free energy**

$$F(R) = H(R) - T S(R).$$

What it is: Theorem 4.1. Lyapunov-type free energy.

31. **Monotone free-energy decay**

$$\frac{dF}{dt} = -T g^{\mu\nu} \partial_\mu S \partial_\nu S \leq 0.$$

What it is: Theorem 4.1. Free energy is non-increasing along metriplectic flow.

32. **Expanded free-energy derivative used in the proof**

$$\frac{dF}{dt} = \frac{dH}{dt} - T \frac{dS}{dt} = (H, S)_M - T g^{\mu\nu} \partial_\mu S \partial_\nu S = -T g^{\mu\nu} \partial_\mu S \partial_\nu S.$$

What it is: The proof chain for Theorem 4.1.

33. **VDM state-variable form**

$$\dot{x} = \{x, H\}_J + (x, S)_M.$$

What it is: Section 4.3. VDM form of the metriplectic evolution law.

34. **Degeneracy verification for entropy**

$$J \cdot \delta S = \Omega^{\mu\nu} \partial_\nu S = 0.$$

What it is: Section 4.3 / 8.1. Explicit J-limb degeneracy check.

35. **Degeneracy verification for energy**

$$M \cdot \delta H = g^{\mu\nu} \partial_\nu H = 0.$$

What it is: Section 4.3. Explicit M-limb degeneracy check.

36. **Finite-difference derivative used in the computation algorithm**

$$|\partial_\mu \psi\rangle \approx \frac{|\psi(R + \delta R^\mu)\rangle - |\psi(R)\rangle}{\delta R^\mu}.$$

What it is: Section 5.1. Numerical derivative formula used to compute the QGT.

37. **Gauge-fixed derivative**

$$|\partial_\mu \psi\rangle_\perp = |\partial_\mu \psi\rangle - \langle \psi | \partial_\mu \psi \rangle |\psi\rangle.$$

What it is: Section 5.1. Parallel-transport / horizontal projection used in the algorithm.

38. **QGT from the projected derivative**

$$Q_{\mu\nu} = \langle \partial_\mu \psi | \partial_\nu \psi \rangle_\perp.$$

What it is: Section 5.1. Computational form of the QGT.

39. **Hermiticity check**

$$(Q_{\mu\nu})^* = Q_{\nu\mu}.$$

What it is: Section 5.2. Numerical/hermiticity verification condition.

40. **Antisymmetry check for Berry curvature**

$$\Omega_{\mu\nu} = -\Omega_{\nu\mu}.$$

What it is: Section 5.2. Structural verification for the curvature sector.

41. **Symmetry check for quantum metric**

$$g_{\mu\nu} = g_{\nu\mu}.$$

What it is: Section 5.2. Structural verification for the metric sector.

42. **Positive-definiteness check for the metric**

$$\lambda_i(g) > 0.$$

What it is: Section 5.2. Eigenvalue test for the quantum metric.

43. **Classical limit of the quantum metric**

$$g_{\mu\nu} \rightarrow g_{\mu\nu}^{\text{cl}} = \partial_\mu q_i m_{ij} \partial_\nu q_j.$$

What it is: Theorem 6.1. Semiclassical limit of the metric sector.

44. **Classical limit of the Berry curvature**

$$\Omega_{\mu\nu} \rightarrow \Omega_{\mu\nu}^{\text{cl}} = \partial_\mu q_i \omega_{ij} \partial_\nu q_j.$$

What it is: Theorem 6.1. Semiclassical limit of the curvature sector.

45. **WKB ansatz**

$$|\psi\rangle = e^{iS(q,R)/\hbar} |\phi(q,R)\rangle.$$

What it is: Section 6.1. Starting point of the semiclassical expansion.

46. **Semiclassical QGT expansion**

$$Q_{\mu\nu} = Q_{\mu\nu}^{(0)} + \hbar Q_{\mu\nu}^{(1)} + O(\hbar^2).$$

What it is: Section 6.1. Expansion used in the classical-limit argument.

47. **Continuum-limit metriplectic evolution**

$$\dot{R}^\mu = \Omega^{\mu\nu} \partial_\nu H + g^{\mu\nu} \partial_\nu S.$$

What it is: Section 6.2. Continuous metriplectic flow recovered in the continuum limit.

48. **Worked example Hamiltonian (two-level system)**

$$H(\mathbf{B}) = -\mathbf{B} \cdot \boldsymbol{\sigma} = -B_x \sigma_x - B_y \sigma_y - B_z \sigma_z.$$

What it is: Section 7.1. Two-level Hamiltonian used in the worked example.

49. **Worked example ground state**

$$|\psi(\mathbf{B})\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle.$$

What it is: Section 7.1. Ground-state representative on the Bloch sphere.

50. **Worked example spherical-coordinate relations**

$$\frac{B_z}{|B|} = \cos \theta, \quad \tan \phi = \frac{B_y}{B_x}.$$

What it is: Section 7.1. Coordinate relations used in the Bloch-sphere parameterization.

51. **Worked example Berry connection**

$$A_\theta = 0, \quad A_\phi = \cos \theta.$$

What it is: Section 7.2. Berry connection components for the two-level system.

52. **Worked example Berry curvature**

$$\Omega_{\theta\phi} = -\sin \theta.$$

What it is: Section 7.2. Berry curvature on the Bloch sphere.

53. **Worked example symplectic form**

$$\omega = \sin \theta d\theta \wedge d\phi.$$

What it is: Section 7.2. The standard area form on the Bloch sphere.

54. **Worked example metric components**

$$g_{\theta\theta} = \frac{1}{4}, \quad g_{\phi\phi} = \frac{1}{4} \sin^2 \theta, \quad g_{\theta\phi} = 0.$$

What it is: Section 7.3. Quantum metric components for the two-level system.

55. **Worked example line element**

$$ds^2 = \frac{1}{4} (d\theta^2 + \sin^2 \theta d\phi^2).$$

What it is: Section 7.3. Metric line element for the Bloch sphere with radius 1/2.

56. **Worked example J-bracket flow**

$$\{\theta, H\}_J = -\frac{\partial H}{\partial \phi} \cdot \frac{1}{\sin \theta}.$$

What it is: Section 7.4. Conservative angular evolution in the worked example.

57. **Worked example M-bracket flow**

$$(\theta, S)_M = 4 \frac{\partial S}{\partial \theta}.$$

What it is: Section 7.4. Dissipative angular evolution in the worked example.

58. **Worked example combined evolution**

$$\dot{\theta} = -\frac{1}{\sin \theta} \frac{\partial H}{\partial \phi} + 4 \frac{\partial S}{\partial \theta}.$$

What it is: Section 7.4. Full metriplectic angular flow in the worked example.

59. **Validation check: entropy as J-Casimir**

$$\{S, H\}_J = \Omega^{\mu\nu} \partial_\mu S \partial_\nu H = 0.$$

What it is: Section 8.1. Explicit verification target for entropy degeneracy on symplectic leaves.

60. **Validation check: energy as M-Casimir**

$$(H, S)_M = g^{\mu\nu} \partial_\mu H \partial_\nu S = 0.$$

What it is: Section 8.1. Explicit verification target for metric degeneracy.

61. **Numerical antisymmetry gate**

$$|\Omega_{\mu\nu} + \Omega_{\nu\mu}| \leq 10^{-12}.$$

What it is: Section 8.2. Machine-precision antisymmetry gate.

62. **Numerical symmetry gate**

$$|g_{\mu\nu} - g_{\nu\mu}| \leq 10^{-12}.$$

What it is: Section 8.2. Machine-precision symmetry gate.

63. **Metric eigenvalue gate**

$$\lambda_{\min}(g) \geq 10^{-10}.$$

What it is: Section 8.2. Positive-definiteness/PSD gate for the metric.

64. **Identity residual gates**

$$|\{\Sigma, H\}_J| \leq 10^{-12}, \quad |(I, S)_M| \leq 10^{-12}.$$

What it is: Section 8.2. Residual tolerances for the degeneracy identities.

65. **Lyapunov monotonicity gate**

$$\frac{dF}{dt} \leq 10^{-12}.$$

What it is: Section 8.2. Numerical gate for monotonic free-energy behavior.

66. **Registry form of the Berry curvature equation**

$$\Omega_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

What it is: Section 9.2 / VDM-E-139. Registry form of the curvature equation.

67. **Registry form of the J-bracket**

$$\{f, g\}_J = \Omega^{\mu\nu} \partial_\mu f \partial_\nu g.$$

What it is: Section 9.2 / VDM-E-141. Registry form of the conservative bracket.

68. **Registry form of the M-bracket**

$$(f, g)_M = g_{\mu\nu} \partial^\mu f \partial^\nu g.$$

What it is: Section 9.2 / VDM-E-142. Registry form of the dissipative bracket as listed in the equation registry.

69. **Topological quantization remark**

$$\frac{1}{2\pi} \int_S \Omega \in \mathbb{Z}.$$

What it is: Section 9.4 remark. Integer quantization of integrated Berry curvature over a closed surface.

70. **Minimal nontrivial winding in the Bloch-sphere example**

$$\int_S \Omega = 2\pi.$$

What it is: Section 9.4 remark. The worked example carries Chern number 1.

4 CF02 - Contact to Metriplectic Evolution

1. **Target metriplectic evolution law**

$$\dot{q} = J(q) \frac{\delta \mathcal{I}}{\delta q} + M(q) \frac{\delta \Sigma}{\delta q}.$$

What it is: The main target structure stated in the objective and abstract.

2. **GENERIC degeneracy constraints**

$$J \frac{\delta \Sigma}{\delta q} = 0, \quad M \frac{\delta \mathcal{I}}{\delta q} = 0.$$

What it is: The defining degeneracy constraints for the metriplectic split.

3. **Contact condition**

$$\alpha \wedge (d\alpha)^n \neq 0.$$

What it is: Definition of a contact manifold.

4. **Reeb defining equations**

$$\iota_R \alpha = 1, \quad \iota_R d\alpha = 0.$$

What it is: Definition of the Reeb vector field.

5. **Reeb splitting of the tangent bundle**

$$TM = \langle R \rangle \oplus \xi.$$

What it is: Pointwise splitting induced by the Reeb field and contact distribution.

6. Darboux contact form

$$\alpha = ds - p_i dq^i.$$

What it is: Standard local contact form in Darboux coordinates.

7. Darboux curvature two-form

$$d\alpha = dq^i \wedge dp_i.$$

What it is: Exterior derivative of the Darboux contact form.

8. Darboux Reeb field

$$R = \frac{\partial}{\partial s}.$$

What it is: Reeb field in Darboux coordinates.

9. Thermodynamic contact form

$$\alpha = dU - T dS - p dV.$$

What it is: First-law contact form used for equilibrium thermodynamics.

10. Legendre-submanifold condition

$$\alpha|_L = 0.$$

What it is: Defining condition for a Legendre submanifold.

11. Equivalent Legendre tangent condition

$$TL \subset \xi.$$

What it is: Equivalent geometric form of the Legendre-submanifold condition.

12. Lagrangian restriction condition

$$d\alpha|_{TL} = 0.$$

What it is: The Legendre submanifold is Lagrangian inside $(\xi, d\alpha|_\xi)$.

13. Contact Hamiltonian defining equations

$$\iota_{X_K} d\alpha = dK - (RK) \alpha, \quad \alpha(X_K) = -K.$$

What it is: Definition of the contact Hamiltonian vector field.

14. Contact Hamiltonian vector field in Darboux coordinates

$$X_K = \frac{\partial K}{\partial p_i} \frac{\partial}{\partial q^i} - \left(\frac{\partial K}{\partial q^i} + p_i \frac{\partial K}{\partial s} \right) \frac{\partial}{\partial p_i} + \left(p_i \frac{\partial K}{\partial p_i} - K \right) \frac{\partial}{\partial s}.$$

What it is: Explicit contact Hamiltonian vector field in Darboux coordinates.

15. **Contact/Jacobi bracket**

$$\{f, g\}_c = d\alpha(X_f, X_g) + f R(g) - g R(f).$$

What it is: Canonical Jacobi bracket carried by a contact manifold.

16. **Poisson bracket from a Poisson operator**

$$\{A, B\} = \left\langle \frac{\delta A}{\delta q}, J \frac{\delta B}{\delta q} \right\rangle.$$

What it is: Antisymmetric bracket used in the metriplectic split.

17. **Metric bracket from a metric operator**

$$(A, B) = \left\langle \frac{\delta A}{\delta q}, M \frac{\delta B}{\delta q} \right\rangle.$$

What it is: Symmetric bracket used in the metriplectic split.

18. **Invariant and entropy evolution laws**

$$\frac{d\mathcal{I}}{dt} = 0, \quad \frac{d\Sigma}{dt} = (\Sigma, \Sigma) \geq 0.$$

What it is: Consequences of the degeneracy constraints.

19. **Poisson operator from the contact two-form**

$$J \equiv \omega^{-1}, \quad \omega = d\alpha.$$

What it is: Construction of the Poisson operator on a nondegenerate symplectic slice.

20. **Darboux Poisson matrix on the (q, p) -subspace**

$$J_{(q,p)} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}.$$

What it is: Canonical symplectic Poisson matrix on the slice.

21. **Raw contravariant metric operator**

$$M_0 \equiv g^{-1}.$$

What it is: Contravariant operator induced from a covariant metric g .

22. **Projector orthogonal to the invariant gradient**

$$P_{\mathcal{I}} = I - \frac{\nabla \mathcal{I} \nabla \mathcal{I}^\top}{\|\nabla \mathcal{I}\|^2}.$$

What it is: Projection removing the $\nabla \mathcal{I}$ component.

23. Projected metric operator

$$M = P_{\mathcal{I}} M_0 P_{\mathcal{I}}.$$

What it is: Constructed metric operator satisfying invariant degeneracy by design.

24. Projected invariant degeneracy

$$M \nabla \mathcal{I} = 0.$$

What it is: Immediate consequence of the projection construction.

25. Entropy choice compatible with J -degeneracy

$$\Sigma = s.$$

What it is: On a state space where J acts only on (q, p) , choosing entropy equal to the contact potential coordinate satisfies the Poisson degeneracy condition.

26. Worked example contact form

$$\alpha = ds - p dq.$$

What it is: Contact form for the worked (q, p, s) example.

27. Worked example curvature two-form

$$d\alpha = dq \wedge dp.$$

What it is: Exterior derivative of the worked-example contact form.

28. Worked example Poisson matrix

$$J = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

What it is: Poisson operator used in the worked example.

29. Worked example invariant generator

$$\mathcal{I}(q, p, s) = \frac{1}{2} (p^2 + \omega^2 q^2).$$

What it is: Energy-like invariant for the worked example.

30. Worked example entropy generator

$$\Sigma(q, p, s) = s.$$

What it is: Entropy-like generator for the worked example.

31. **Worked example contact metric**

$$g = dq^2 + dp^2 + \alpha^2 = dq^2 + dp^2 + (ds - p dq)^2.$$

What it is: Standard contact metric chosen in the worked example.

32. **Worked example covariant metric matrix**

$$[g] = \begin{bmatrix} 1 + p^2 & 0 & -p \\ 0 & 1 & 0 \\ -p & 0 & 1 \end{bmatrix}.$$

What it is: Matrix of the worked-example contact metric in the (q, p, s) basis.

33. **Worked example determinant**

$$\det[g] = 1.$$

What it is: Determinant of the worked-example metric matrix.

34. **Worked example raw contravariant metric operator**

$$[M_0] = [g]^{-1} = \begin{bmatrix} 1 & 0 & p \\ 0 & 1 & 0 \\ p & 0 & 1 + p^2 \end{bmatrix}.$$

What it is: Inverse metric used as the raw metric operator.

35. **Worked example parameter specialization**

$$\omega = 2.$$

What it is: Parameter value used in the numerical validation sweep.

36. **Worked example validation domain**

$$q, p \in [-1, 1].$$

What it is: Numerical domain used for the validation sweep.

37. **Worked example grid resolution**

$$51 \times 51.$$

What it is: Grid size used for the deterministic sweep.

38. **Worked example fixed-potential condition**

$$s = \text{constant}.$$

What it is: The validation sweep is performed at fixed contact-potential coordinate.

39. Worked example Poisson degeneracy

$$J \nabla \Sigma = 0.$$

What it is: Since $\Sigma = s$ and J has no s -components, the Poisson degeneracy holds identically.

40. Symmetry gate for the metric operator

$$\|M - M^\top\|_F \leq 10^{-10}.$$

What it is: Publishable gate for symmetry of the constructed metric operator.

41. Observed symmetry residual

$$\|M - M^\top\|_F = 1.9 \times 10^{-16}.$$

What it is: Observed value from the worked-example sweep.

42. PSD gate for the metric operator

$$\lambda_{\min}(M) \geq -10^{-10}.$$

What it is: Publishable gate for positive semidefiniteness of the metric operator.

43. Observed minimum eigenvalue

$$\lambda_{\min}(M) = -4.8 \times 10^{-16}.$$

What it is: Observed minimum eigenvalue from the worked-example sweep.

44. Invariant-degeneracy gate

$$\|M \nabla \mathcal{I}\| \leq 10^{-10}.$$

What it is: Publishable degeneracy gate for invariant preservation.

45. Observed invariant-degeneracy residual

$$\|M \nabla \mathcal{I}\| = 1.1 \times 10^{-15}.$$

What it is: Observed degeneracy residual from the worked-example sweep.

46. Entropy-production gate

$$\dot{\Sigma} \geq -10^{-12}.$$

What it is: Publishable entropy-production sign gate.

47. **Observed minimum entropy production**

$$\dot{\Sigma}_{\min} = 1.0.$$

What it is: Observed minimum entropy production in the sweep.

48. **Poisson-sector antisymmetry gate**

$$J^\top = -J.$$

What it is: Minimal publishable gate for the Poisson sector.

49. **Jacobi gate**

$$\|\text{Jacobi}(J)\| \leq \varepsilon.$$

What it is: Decisive falsifier/gate for the Poisson sector.

50. **Metric-sector symmetry gate**

$$M^\top = M.$$

What it is: Minimal publishable gate for the metric sector.

51. **Metric-sector positive-semidefinite gate**

$$M \succeq 0.$$

What it is: Minimal publishable PSD condition for the metric sector.

52. **Entropy law validation gate**

$$\dot{\Sigma} = (\Sigma, \Sigma)_M \geq 0.$$

What it is: Final entropy-law gate summarized at the end of the paper.

5 CF03 - A8 Scaling Theorem (Hierarchical Tachyonic Interfaces)

1. **A8 scaling claim**

$$N(L) = \Theta\left(\log\left(\frac{L}{\ell_0}\right)\right).$$

What it is: The central scaling claim formalized in the objective and abstract.

2. **Minimal phase-field energy**

$$E_\varepsilon[\phi] = \int_\Omega \left[\frac{\varepsilon}{2} |\nabla \phi|^2 + \frac{1}{\varepsilon} W(\phi) \right] dx.$$

What it is: The canonical tachyonic/interface-forming phase-field energy.

3. Sharp-interface Γ -limit

$$E_\varepsilon \xrightarrow{\Gamma} E_0 \equiv c_0 \operatorname{Per}(A), \quad A \equiv \{x \in \Omega : \phi(x) \approx +1\}.$$

What it is: Claim C1. The Modica–Mortola sharp-interface bridge.

4. Surface-tension constant

$$c_0 = \int_{-1}^1 \sqrt{2W(s)} \, ds.$$

What it is: The interface-energy constant appearing in the Γ -limit.

5. Boundary-energy concentration / surface-dominated limit

$$E_{\text{exc}} \propto \operatorname{Per}(A).$$

What it is: Claim C2 in condensed formula form: excess energy concentrates on the interface.

6. Logarithmic upper bound on active depth

$$N(L) \leq \left\lceil \log_2 \left(\frac{L}{\ell_0} \right) \right\rceil + 1.$$

What it is: Claim C3. The universal logarithmic upper bound on active hierarchy depth.

7. Lower realizability witness for log-depth hierarchy

$$N(L) \geq c \log \left(\frac{L}{\ell_0} \right), \quad c > 0.$$

What it is: Claim C3. Hierarchical constructions can realize logarithmic depth from below.

8. No per-scale combinatorial blow-up target

$$m_k \not\sim 2^k.$$

What it is: Claim C4 in compact form. Active interfaces should not grow exponentially with dyadic level under perimeter-reducing dynamics.

9. Perimeter-proxy scaling gate

$$E_{\text{exc}} \approx \sigma \widehat{\operatorname{Per}}.$$

What it is: Gate G1. The discrete perimeter proxy must linearly explain excess energy in the sharp-interface regime.

10. **Perimeter-proxy relative-error gate**

$$\frac{|E_{\text{exc}} - \sigma \widehat{\text{Per}}|}{E_{\text{exc}}} \leq 0.05.$$

What it is: Gate G1 threshold written as a formula.

11. **Bounded active multiplicity per scale**

$$m_k \leq m_{\text{max}}, \quad m_{\text{max}} \text{ independent of } k.$$

What it is: Gate G2. Active interface multiplicity per scale remains uniformly bounded.

12. **Depth model-selection fit**

$$N(L) = a \log\left(\frac{L}{\ell_0}\right) + b.$$

What it is: Gate G3. The target fitted model for active depth.

13. **Depth model-selection goodness-of-fit gate**

$$R^2 \geq 0.98.$$

What it is: Gate G3. Required fit quality for the logarithmic model.

14. **Dimensionless-collapse variable**

$$\frac{L}{\ell_0}.$$

What it is: Gate G4. The dimensionless scaling variable on which curves should collapse.

15. **Tachyonic / double-well instability condition**

$$W''(0) < 0.$$

What it is: The unstable-origin condition defining the tachyonic/double-well regime.

16. **Double-well minima**

$$W(\pm 1) = 0.$$

What it is: The stable well condition for the phase-field potential.

17. **Strict positivity away from the wells**

$$W(s) > 0 \quad \text{for } s \neq \pm 1.$$

What it is: The nondegeneracy condition of the double-well potential.

18. **Standard double-well example**

$$W(s) = \frac{1}{4}(1 - s^2)^2.$$

What it is: The canonical quartic double-well potential used as the standard example.

19. **Microscopic cutoff relation**

$$\ell_0 \sim \max\{\varepsilon, \Delta x\}.$$

What it is: The effective microscopic cutoff in lattice realizations.

20. **VDM scalar-field special-case evolution law**

$$\dot{q} = J \frac{\delta \mathcal{I}}{\delta q} + M \frac{\delta \Sigma}{\delta q}.$$

What it is: The schematic metriplectic evolution law to which the scalar-field specialization is connected.

21. **Γ -convergence liminf inequality**

$$\liminf_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon) \geq F_0(u) \quad \text{whenever } u_\varepsilon \rightarrow u.$$

What it is: Part (i) of the informal Γ -convergence definition.

22. **Γ -convergence limsup / recovery-sequence inequality**

$$\limsup_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon) \leq F_0(u) \quad \text{for some recovery sequence } u_\varepsilon \rightarrow u.$$

What it is: Part (ii) of the informal Γ -convergence definition.

23. **Modica–Mortola limit functional**

$$E_0[\chi_A] = c_0 \operatorname{Per}(A).$$

What it is: Theorem 3.1. The sharp-interface energy in the limit.

24. **Interface-area scaling**

$$E_{\text{exc}} \approx c_0 \operatorname{Per}(A) = \Theta(L^{d-1}).$$

What it is: Equation (??). Surface-energy scaling for bounded-shape interfaces.

25. **Representative multiscale perimeter sum**

$$\operatorname{Per}(A) \sim \sum_{k=0}^{N(L)-1} L^{d-1} 2^{-k(d-1)}.$$

What it is: The multiscale perimeter model used to show compatibility between log-depth hierarchy and surface scaling.

26. **Geometric-series perimeter bound**

$$\text{Per}(A) \leq \frac{1}{1 - 2^{-(d-1)}} L^{d-1} \quad (d \geq 2).$$

What it is: Equation (??). The total perimeter remains surface-order even with logarithmic depth.

27. **Dyadic scale ladder**

$$\ell_k \equiv 2^{-k} L, \quad k = 0, 1, 2, \dots$$

What it is: Definition of the discrete scale ladder used to define active depth.

28. **Maximum dyadic level**

$$K_{\max}(L) = \left\lfloor \log_2 \left(\frac{L}{\ell_0} \right) \right\rfloor.$$

What it is: The largest dyadic level whose scale is still above the microscopic cutoff.

29. **Active hierarchy depth**

$$N(L) = \# \{k \in \{0, \dots, K_{\max}(L)\} : \ell_k \text{ is active}\}.$$

What it is: Definition of the active hierarchy depth.

30. **Immediate counting bound on active depth**

$$N(L) \leq K_{\max}(L) + 1.$$

What it is: Proposition 5.1. The direct dyadic-count upper bound.

31. **Allen–Cahn gradient flow**

$$\partial_t \phi = \varepsilon \Delta \phi - \frac{1}{\varepsilon} W'(\phi).$$

What it is: The perimeter-reducing dissipative model used to justify suppression of same-scale multiplicity.

32. **Gradient perimeter proxy**

$$\widehat{\text{Per}} \equiv \sum_x |\nabla \phi(x)| \Delta x^d.$$

What it is: First discrete perimeter proxy defined for the CFN extraction protocol.

33. Binary threshold field

$$\chi(x) = \mathbf{1}_{\phi(x) > 0}.$$

What it is: Binary phase field used for the sign-change perimeter proxy.

34. Scale-by-scale coarse-grained field

$$\phi_k \equiv \mathcal{C}_{\ell_k}[\phi].$$

What it is: Coarse-grained field at dyadic scale ℓ_k .

35. Boundary extraction at scale ℓ_k

$$\mathcal{B}_k \equiv \mathcal{B}(\phi_k; \ell_k).$$

What it is: The extracted visible boundary set at scale ℓ_k .

36. New-boundary criterion

$$\Delta \mathcal{B}_k \equiv \mathcal{B}_k \setminus \text{dilate}(\mathcal{B}_{k-1}).$$

What it is: Definition used to avoid double-counting interfaces across scales.

37. Depth from the active-boundary census

$$N(L) = \#\{k : m_k > 0\}.$$

What it is: Operational form of active depth using per-scale multiplicities m_k .

38. Log-fit model for depth across domain-size sweep

$$N(L) = a \log\left(\frac{L}{\ell_0}\right) + b.$$

What it is: The explicit fitted law used in CFN model selection.

39. Power-law alternative for model comparison

$$N(L) = c \left(\frac{L}{\ell_0}\right)^\alpha.$$

What it is: The main competing model rejected if information criteria favor the logarithmic law.

40. Worked example surface-tension constant

$$c_0 = \int_{-1}^1 \sqrt{\frac{1}{2}(1-s^2)^2} ds = \frac{2\sqrt{2}}{3}.$$

What it is: Explicit evaluation of the surface-tension constant for the standard quartic double well.

41. **One-dimensional sharp-interface energy**

$$E_0 = c_0 \operatorname{Per}(A).$$

What it is: In $d = 1$, sharp-interface energy counts boundary points times c_0 .

42. **One-dimensional log-depth witness**

$$N(L) \approx \left\lfloor \log_2 \left(\frac{L}{\ell_0} \right) \right\rfloor + 1.$$

What it is: The nested dyadic construction realizing logarithmic hierarchy depth in 1D.

43. **One-dimensional energy of the nested hierarchy**

$$E_0 \approx c_0 N(L) = \Theta \left(\log \left(\frac{L}{\ell_0} \right) \right).$$

What it is: The worked-example energy scaling for a log-depth hierarchy in one dimension.

6 CF04 - Telegraph–Fisher Causality

1. **Continuity law**

$$\partial_t u + \nabla \cdot J = 0.$$

What it is: Conservation law for the transported scalar density.

2. **Cattaneo–Vernotte flux relaxation law**

$$\tau \partial_t J + J = -D \nabla u, \quad D > 0, \tau > 0.$$

What it is: Finite-relaxation constitutive replacement for instantaneous Fickian transport.

3. **Telegrapher equation**

$$\tau \partial_{tt} u + \partial_t u = D \Delta u.$$

What it is: Derived by eliminating the flux J from the continuity law and the Cattaneo–Vernotte closure.

4. **Characteristic speed**

$$c \equiv \sqrt{\frac{D}{\tau}}.$$

What it is: The finite causal speed of the telegrapher equation.

5. **Plane-wave ansatz**

$$u \sim e^{i(k \cdot x - \omega t)}.$$

What it is: Mode ansatz used to derive the dispersion relation.

6. **Telegraph dispersion polynomial**

$$\tau \omega^2 + i \omega - D |k|^2 = 0.$$

What it is: Characteristic polynomial for the telegraph equation under the plane-wave ansatz.

7. **Telegraph dispersion relation**

$$\omega_{\pm}(k) = \frac{i}{2\tau} \pm \sqrt{\frac{D}{\tau} |k|^2 - \frac{1}{4\tau^2}}.$$

What it is: The two telegraph branches obtained from the dispersion polynomial.

8. **Telegraph–Fisher/KPP equation**

$$\tau u_{tt} + u_t = D u_{xx} + r u(1 - u), \quad r > 0.$$

What it is: Telegraph transport augmented with a logistic reaction source.

9. **Minimum front speed law**

$$v_{\min}(\tau) = 2 \sqrt{\frac{Dr}{1 + 4r\tau}}.$$

What it is: Minimum traveling-wave speed for the constant-damping telegraph–Fisher/KPP model cited in the paper.

10. **Small- $r\tau$ front-speed expansion**

$$v_{\min}(\tau) = 2\sqrt{Dr} (1 - 2r\tau + \mathcal{O}((r\tau)^2)).$$

What it is: The small- κ expansion of the minimum front speed.

11. **Fisher information**

$$\mathcal{I}_F[p] \equiv \int \frac{|\nabla p|^2}{p} dx.$$

What it is: The Fisher-information functional used in the epistemic-causality interpretation.

12. **Cramér–Rao bound**

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}_F}.$$

What it is: The variance lower bound cited in the Fisher-information interpretation section.

13. **Finite-speed support theorem**

$$\text{supp}(u(\cdot, t)) \subseteq B_{R_0+ct}, \quad c = \sqrt{D/\tau}.$$

What it is: Theorem 4.1 in formula form. Compactly supported initial data remain inside the forward causal cone.

14. **Exponential damping substitution**

$$u(x, t) = e^{-t/(2\tau)} w(x, t).$$

What it is: The standard substitution used to rewrite the telegraph equation as a damped wave-type equation.

15. **Wave-type equation after damping removal**

$$\tau w_{tt} = D \Delta w + \frac{1}{4\tau} w.$$

What it is: The transformed equation for w , used in the proof sketch for finite propagation speed.

16. **Diffusion equation**

$$\partial_t u = D \Delta u.$$

What it is: The parabolic diffusion limit recovered as $\tau \rightarrow 0$.

17. **Long-wavelength slow branch**

$$\omega \approx -iD|k|^2 + \mathcal{O}(\tau|k|^4).$$

What it is: The diffusive asymptotic branch of the telegraph dispersion relation for $|k| \ll 1/\sqrt{D\tau}$.

18. **Fast transient mode**

$$\Im \omega \approx +\frac{1}{\tau}.$$

What it is: The large-decay-rate transient branch in the long-wavelength regime, as stated in the paper.

19. **Short-wavelength damped-wave asymptotics**

$$\omega \approx \pm c|k| - \frac{i}{2\tau}.$$

What it is: The short-wavelength asymptotic form of the telegraph dispersion relation.

20. **Causal cap on the front speed**

$$v_{\min}(\tau) < c = \sqrt{D/\tau}.$$

What it is: Proposition 6.1. Telegraph–Fisher fronts remain below the causal speed.

21. **Dimensionless control parameter**

$$\kappa \equiv r\tau.$$

What it is: The dimensionless parameter controlling deviation from the diffusive Fisher limit.

22. **Observed support radius**

$$R_\epsilon(t) = \sup\{|x| : |u(x, t)| \geq \epsilon\}.$$

What it is: Operational support-radius observable used in Gate G1.

23. **Finite-support-growth validation gate**

$$R_\epsilon(t) \leq R_0 + ct + \delta_R.$$

What it is: Gate G1. Numerical pass condition for finite propagation speed.

24. **Dispersion-fit validation gate**

$$\text{relative fit error} \leq 5\%.$$

What it is: Gate G2 threshold for fitting measured $(\Re\omega(k), \Im\omega(k))$ to the telegraph dispersion law.

25. **Front-speed scaling validation gate**

$$\frac{|v_{\text{meas}} - v_{\min}(\tau)|}{v_{\min}(\tau)} \leq 5\%.$$

What it is: Gate G3 threshold for the telegraph–Fisher front-speed law across a τ -sweep.

26. **Causal-cap validation condition on measured fronts**

$$v_{\text{meas}} < c.$$

What it is: Additional Gate G3 condition requiring measured fronts to remain subcausal.

27. **First-order form: state-velocity split**

$$u_t = v.$$

What it is: First equation in the first-order formulation of the telegraph dynamics.

28. **First-order form: damped forcing equation**

$$\tau v_t = D \Delta u - v.$$

What it is: Second equation in the first-order formulation of the telegraph dynamics.

29. **Canonical VDM speed scale hook**

$$c = \sqrt{D/\tau}.$$

What it is: The reusable canon speed scale identified in the integration map.

30. **Effective throttled diffusivity hook**

$$D_{\text{eff}} = \frac{D_0}{1 + \tau D_0 k^2}.$$

What it is: The frequency/wavenumber-dependent attenuation form referenced in the canon integration section.

7 CF05 - Integrability Closure (No Hidden Conserved Quantities)

1. **GENERIC / metriplectic evolution law**

$$\dot{x} = L(x) \nabla E(x) + M(x) \nabla S(x).$$

What it is: The base metriplectic evolution equation used throughout the closure analysis.

2. **Poisson bracket**

$$\{F, G\}_J = \nabla F^\top L \nabla G.$$

What it is: The antisymmetric bracket generated by the Poisson operator L .

3. **Degeneracy identities**

$$L \nabla S = 0, \quad M \nabla E = 0.$$

What it is: The structural GENERIC degeneracy constraints used to separate invariant and entropy roles.

4. **Energy conservation law**

$$\frac{dE}{dt} = 0.$$

What it is: Structural invariant law claimed in C1.

5. **Entropy production law**

$$\frac{dS}{dt} = \nabla S^\top M \nabla S \geq 0.$$

What it is: Structural entropy law claimed in C1.

6. **Expanded energy-balance identity**

$$\frac{dE}{dt} = \nabla E^\top L \nabla E + \nabla E^\top M \nabla S = 0.$$

What it is: The explicit calculation showing structural energy conservation from antisymmetry and degeneracy.

7. **Expanded entropy-balance identity**

$$\frac{dS}{dt} = \nabla S^\top L \nabla E + \nabla S^\top M \nabla S = \nabla S^\top M \nabla S \geq 0.$$

What it is: The explicit calculation showing structural entropy production from degeneracy and positive semidefiniteness.

8. **First-integral condition**

$$\frac{d}{dt} I(x(t)) = \nabla I(x)^\top F(x) = 0.$$

What it is: Definition of a first integral for an autonomous system $\dot{x} = F(x)$.

9. **Full-flow hidden-invariant condition**

$$\nabla I \cdot (L \nabla E + M \nabla S) = 0.$$

What it is: Condition for an unintended first integral of the full metriplectic flow.

10. **Poisson Casimir condition**

$$L \nabla C = 0.$$

What it is: Condition for a hidden invariant of the reversible limb.

11. **Metric-null invariant condition**

$$M \nabla J = 0.$$

What it is: Condition for a hidden invariant of the irreversible limb.

12. **Jacobi validity gate**

$$e_{\text{Jacobi}} \leq 10 \varepsilon_{\text{mach}}.$$

What it is: Gate G0. Integrability closure is not meaningful unless the Poisson structure satisfies Jacobi to machine precision on the declared basis.

13. **Degeneracy correctness gate**

$$\|L \nabla S\|_\infty \leq 10 \varepsilon_{\text{mach}}, \quad \|M \nabla E\|_\infty \leq 10 \varepsilon_{\text{mach}}.$$

What it is: Gate G1. Structural GENERIC correctness requirement.

14. **Entropy-production gate**

$$\Delta S \geq -10 \varepsilon_{\text{mach}}$$

per step and cumulatively. *What it is:* Gate G2. Numerical entropy must remain nonnegative up to machine tolerance.

15. **Candidate invariant drift functional**

$$\text{drift}(I) := \max_{t \in [0, T]} \frac{|I(x(t)) - I(x(0))|}{\max(1, |I(x(0))|)}.$$

What it is: Gate G4. Relative drift measure used to verify any discovered candidate invariant.

16. **Candidate invariant drift gate**

$$\text{drift}(I) \leq 10^{-8}.$$

What it is: Gate G4 threshold across declared seeds and step-size refinements.

17. **Darboux polynomial condition**

$$\nabla f(x)^\top F(x) = K(x) f(x).$$

What it is: Definition of a Darboux polynomial with cofactor $K(x)$.

18. **Darbouxian first-integral ansatz**

$$I(x) = \prod_i f_i(x)^{\lambda_i}.$$

What it is: Candidate invariant constructed from Darboux polynomials with compatible cofactors.

19. **Prelle–Singer exactness target**

$$\mu(x) (F(x) \cdot dx)$$

is an exact differential. *What it is:* The integrating-factor target in the Prelle–Singer search family.

20. **Data-driven invariant regression ansatz**

$$I(x) = \sum_j c_j \phi_j(x).$$

What it is: Candidate invariant family used in the numerical discovery phase.

21. **Trajectory-based infinitesimal regression target**

$$\nabla I(x(t_k))^\top F(x(t_k)) \approx 0.$$

What it is: Least-squares target for discovering candidate invariants from trajectory data.

22. **Finite-difference drift proxy**

$$\dot{I}(t_k) \approx \frac{I(t_{k+1}) - I(t_k)}{\Delta t}.$$

What it is: Numerical approximation used in the trajectory-based discovery phase.

23. **Functional-independence filter**

$$\text{rank}[\nabla E, \nabla C_1, \dots, \nabla I]$$

must increase generically. *What it is:* Practical test that a discovered invariant candidate is not functionally dependent on known invariants.

24. **Closure-discovery failure condition**

$$\exists I \text{ independent such that } \text{drift}(I) \leq 10^{-8}.$$

What it is: Gate G3 fails if any additional independent invariant survives discovery and verification over the declared search class.

25. **Worked example: canonical 2D Jacobi residual**

$$e_{\text{Jacobi}} = 2.32 \times 10^{-10}.$$

What it is: Finite-difference nested-bracket Jacobi residual reported by the companion notebook for the canonical 2D Poisson example.

26. **Worked example: Poisson bilinearity residual**

$$5.55 \times 10^{-17}.$$

What it is: Reported bilinearity residual in the minimal sanity-check notebook.

27. **Worked example: Poisson antisymmetry residual**

$$0.$$

What it is: Reported antisymmetry residual in the minimal sanity-check notebook.

28. **Worked example: energy drift under exact symplectic rotation**

$$5.94 \times 10^{-15}.$$

What it is: Reported energy-drift residual for the harmonic-oscillator rotation sanity check.

8 CF06 - Information Geometry Foundations (Fisher and Ruppeiner)

1. Local KL expansion

$$D_{\text{KL}}(p(\cdot \mid \theta) \parallel p(\cdot \mid \theta + d\theta)) = \frac{1}{2} g_{ij}^{\text{F}}(\theta) d\theta^i d\theta^j + \mathcal{O}(\|d\theta\|^3).$$

What it is: Claim C1. Fisher geometry as the intrinsic local quadratic form of statistical distinguishability.

2. Ruppeiner metric

$$g_{ij}^{\text{R}}(X) := -\frac{\partial^2(S/k_B)}{\partial X^i \partial X^j}.$$

What it is: Claim C3. The entropy-Hessian metric on equilibrium thermodynamic state space.

3. Score function

$$\ell_i(x; \theta) := \partial_i \log p(x \mid \theta).$$

What it is: Definition of the score used to build Fisher geometry.

4. Score-centering identity

$$\mathbb{E}_\theta[\ell_i] = \int p \partial_i \log p \, dx = \int \partial_i p \, dx = \partial_i \int p \, dx = 0.$$

What it is: Centering identity for the score under standard regularity assumptions.

5. Fisher information metric

$$g_{ij}^{\text{F}}(\theta) := \mathbb{E}_\theta[\ell_i \ell_j] = \int p(x \mid \theta) \partial_i \log p(x \mid \theta) \partial_j \log p(x \mid \theta) \, dx.$$

What it is: Definition of the Fisher–Rao metric.

6. Equivalent Hessian form of Fisher information

$$g_{ij}^{\text{F}}(\theta) = -\mathbb{E}_\theta[\partial_i \partial_j \log p(x \mid \theta)].$$

What it is: Equivalent Fisher formula under differentiation-under-the-integral regularity assumptions.

7. Positive-semidefiniteness of the Fisher metric

$$v^i g_{ij}^{\text{F}} v^j = \mathbb{E}_\theta[(v^i \ell_i)^2] \geq 0.$$

What it is: The Fisher metric is symmetric PSD.

8. **KL logarithm expansion**

$$\log \frac{p(x \mid \theta)}{p(x \mid \theta + d\theta)} = -\partial_i \log p d\theta^i - \frac{1}{2} \partial_i \partial_j \log p d\theta^i d\theta^j + \mathcal{O}(\|d\theta\|^3).$$

What it is: Expansion used to derive the local KL quadratic form.

9. **Univariate Gaussian family**

$$p(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

What it is: Worked statistical family used to compute a Fisher metric explicitly.

10. **Gaussian score components**

$$\partial_\mu \log p = \frac{x - \mu}{\sigma^2}, \quad \partial_\sigma \log p = -\frac{1}{\sigma} + \frac{(x - \mu)^2}{\sigma^3}.$$

What it is: Score derivatives for the Gaussian family.

11. **Gaussian Fisher matrix**

$$g^F(\mu, \sigma) = \begin{pmatrix} \sigma^{-2} & 0 \\ 0 & 2\sigma^{-2} \end{pmatrix}.$$

What it is: Explicit Fisher matrix for the Gaussian family in (μ, σ) coordinates.

12. **Gaussian Fisher components**

$$g_{\mu\mu} = \sigma^{-2}, \quad g_{\sigma\sigma} = 2\sigma^{-2}, \quad g_{\mu\sigma} = 0.$$

What it is: Component form of the Gaussian Fisher metric.

13. **Log-scale Gaussian metric component**

$$g_{\eta\eta} = 2, \quad \eta = \log \sigma.$$

What it is: Fisher metric component under reparameterization to log-scale width.

14. **Cramér–Rao inequality**

$$\text{Cov}(\hat{\theta}) \succeq \frac{1}{n} (g^F(\theta))^{-1}.$$

What it is: Fundamental precision bound for unbiased estimation.

15. **Weinhold metric**

$$g_{ij}^W := \frac{\partial^2 U}{\partial X^i \partial X^j}.$$

What it is: Energy-Hessian metric in extensive thermodynamic coordinates.

16. Ruppeiner metric in dimensionless entropy form

$$g_{ij}^{\text{R}} := -\frac{\partial^2 s}{\partial X^i \partial X^j}, \quad s := S/k_B.$$

What it is: Dimensionless-entropy form of the Ruppeiner metric.

17. Weinhold–Ruppeiner conformal relation

$$g^{\text{R}} = \frac{1}{T} g^{\text{W}}.$$

What it is: Standard conformal relation between the two thermodynamic metrics in appropriate coordinates.

18. Gaussian fluctuation law from Ruppeiner geometry

$$\mathbb{P}(\Delta X) \propto \exp\left(-\frac{1}{2} g_{ij}^{\text{R}} \Delta X^i \Delta X^j\right).$$

What it is: Local equilibrium fluctuation law controlled by the Ruppeiner metric.

19. Ruppeiner covariance relation

$$\text{cov}(X) \approx (g^{\text{R}})^{-1}.$$

What it is: The inverse Ruppeiner metric approximates fluctuation covariance in the appropriate ensemble.

20. Canonical/exponential-family ensemble

$$p(x \mid \theta) = \exp(\theta^a T_a(x) - \psi(\theta)).$$

What it is: MaxEnt exponential-family form used for the Fisher–Ruppeiner bridge.

21. Score in exponential families

$$\partial_a \log p = T_a - \partial_a \psi.$$

What it is: Score function specialized to the exponential-family form.

22. Fisher metric as covariance in multiplier space

$$g_{ab}^{\text{F}}(\theta) = \text{Cov}_{\theta}(T_a, T_b) = \partial_a \partial_b \psi(\theta).$$

What it is: Fisher geometry on multiplier space for exponential families.

23. Canonical-energy Fisher component

$$g_{\beta\beta}^{\text{F}} = \text{Var}(E).$$

What it is: One-dimensional canonical-ensemble Fisher relation.

24. **Expectation coordinates**

$$\eta_a := \mathbb{E}_\theta[T_a].$$

What it is: Dual expectation-space coordinates for the Fisher–Ruppeiner bridge.

25. **Legendre-dual entropy potential**

$$s(\eta) = \theta^a \eta_a - \psi(\theta).$$

What it is: Dimensionless entropy as the Legendre dual of the log-partition function.

26. **Fisher–Ruppeiner duality**

$$g_{ab}^{\text{R}}(\eta) = -\frac{\partial^2 s}{\partial \eta_a \partial \eta_b} = (g^{\text{F}}(\theta))_{ab}^{-1}.$$

What it is: Claim C4. The explicit bridge between covariance geometry and entropy-Hessian geometry.

27. **One-dimensional canonical bridge**

$$g_{\beta\beta}^{\text{F}} g_{UU}^{\text{R}} = 1.$$

What it is: One-dimensional Fisher–Ruppeiner duality relation in canonical coordinates.

28. **Canonical fluctuation relation**

$$\text{Var}(E) = k_B T^2 C_V.$$

What it is: Standard fluctuation relation used in the canonical bridge.

29. **Canonical Ruppeiner component**

$$g_{UU}^{\text{R}} = \frac{1}{k_B T^2 C_V}.$$

What it is: One-dimensional Ruppeiner component written in measurable thermodynamic quantities.

30. **Natural-gradient flow**

$$\dot{\theta}^i = M^{ij}(\theta) \partial_j \Sigma(\theta), \quad M^{ij}(\theta) := (g^{\text{F}}(\theta))_{ij}^{-1}.$$

What it is: Information-geometric dissipation law as a natural-gradient flow.

31. **Entropy production under the natural-gradient flow**

$$\dot{\Sigma} = \partial_i \Sigma M^{ij} \partial_j \Sigma \geq 0.$$

What it is: Monotone entropy production induced by the inverse Fisher metric.

32. **Metric-induced dissipation bracket**

$$(f, g)_M := \partial_i f M^{ij}(\theta) \partial_j g.$$

What it is: Symmetric bracket form induced by the information metric.

33. **Projected information-geometric metric operator**

$$M = P^\top (g^F)^{-1} P, \quad P := I - \frac{\nabla \mathcal{I} \nabla \mathcal{I}^\top}{\|\nabla \mathcal{I}\|^2}.$$

What it is: Projection construction enforcing invariant degeneracy in the dissipative limb.

34. **Invariant-degeneracy condition**

$$M \nabla \mathcal{I} = 0.$$

What it is: The projected metric operator annihilates the invariant gradient.

35. **KL residual gate**

$$\epsilon_{\text{KL}} \leq 10^{-10}.$$

What it is: Claim C1 decisive metric for the local-KL quadratic approximation.

36. **Monotonicity residual gate under Markov maps**

$$\epsilon_{\text{mono}} \leq 10^{-10}.$$

What it is: Claim C2 decisive metric for Fisher monotonicity under coarse-graining.

37. **Equilibrium fluctuation residual gate**

$$\epsilon_{\text{fluct}} \leq 10^{-8}.$$

What it is: Claim C3 decisive metric for the Ruppeiner fluctuation law.

38. **Fisher–Ruppeiner mismatch gate**

$$\epsilon_{\text{FR}} \leq 10^{-8}.$$

What it is: Claim C4 / Gate G2 decisive threshold for canonical toy-system agreement.

39. **Coordinate-invariance curvature gate**

$$\frac{\max |R - R'|}{\max(|R|, 1)} \leq 10^{-8}.$$

What it is: Gate G1. Scalar curvature (or other scalar invariant) must agree across smooth reparameterizations.

9 CF07 - Measurement Theory Foundations (Decoherence, Einselection, and Born Rule Emergence)

1. Total system–environment Hamiltonian

$$H_{\text{tot}} = H_S \otimes I_E + I_S \otimes H_E + H_{SE}.$$

What it is: The full Hamiltonian governing system–environment coupling.

2. Joint unitary evolution

$$\rho_{SE}(t) = U(t) \rho_{SE}(0) U(t)^\dagger, \quad U(t) = e^{-iH_{\text{tot}}t/\hbar}.$$

What it is: Unitary evolution of the joint system for an initial product state.

3. Reduced system state

$$\rho_S(t) = \text{Tr}_E[\rho_{SE}(t)].$$

What it is: The operational state of the measured system after tracing out the environment.

4. Entangled measurement-channel state

$$|\Psi(t)\rangle = \sum_i c_i |i\rangle \otimes |E_i(t)\rangle.$$

What it is: The standard entangled form induced by system–environment correlation.

5. Reduced state in the record basis

$$\rho_S(t) = \sum_{i,j} c_i c_j^* \langle E_j(t) | E_i(t) \rangle |i\rangle \langle j|.$$

What it is: Reduced density matrix showing suppression of off-diagonals through environment overlaps.

6. Decoherence factor

$$\Gamma_{ij}(t) \equiv \langle E_j(t) | E_i(t) \rangle.$$

What it is: The environment-overlap factor controlling interference suppression.

7. Dephased diagonal state

$$\rho_{\text{diag}} = \sum_i |c_i|^2 |i\rangle \langle i|.$$

What it is: The fully dephased mixture in the candidate pointer basis.

8. **Trace-distance decoherence bound**

$$\frac{1}{2} \|\rho_S(t) - \rho_{\text{diag}}\|_1 \leq \sum_{i < j} |c_i| |c_j| |\Gamma_{ij}(t)|.$$

What it is: Lemma bounding distance to the diagonal state by the decoherence factors.

9. **Uniform decoherence-factor bound**

$$D(t) \leq \gamma(t) \sum_{i < j} |c_i| |c_j| \leq \gamma(t) \frac{d-1}{2}, \quad \max_{i \neq j} |\Gamma_{ij}(t)| \leq \gamma(t).$$

What it is: Simplified trace-distance bound when all off-diagonal overlaps are uniformly bounded.

10. **Einselection / pointer-basis residual**

$$\|[H_{SE}, \Pi_i]\| \leq \varepsilon_{\text{ein}} \quad \text{for all } i.$$

What it is: Operational stability condition defining pointer projectors.

11. **Einselection gate**

$$\max_i \|[H_{SE}, \Pi_i]\| \leq \varepsilon_{\text{ein}}.$$

What it is: Gate G1. The decisive residual for the pointer-basis claim.

12. **Decoherence threshold time**

$$\tau_{\text{dec}}(\varepsilon_{\text{dec}}) \equiv \inf \{ t \geq 0 : D(t) \leq \varepsilon_{\text{dec}} \}.$$

What it is: Operational definition of the decoherence timescale.

13. **Interaction energy scaling estimate**

$$\Delta E \sim \lambda^2 k_B T.$$

What it is: Order-of-magnitude estimate for the effective interaction energy in a thermal environment.

14. **Decoherence-timescale scaling estimate**

$$\tau_{\text{dec}} \sim \frac{\hbar}{\lambda^2 k_B T}.$$

What it is: Model-level scaling estimate for decoherence time.

15. **Decoherence-threshold gate**

$$D(t) \leq \varepsilon_{\text{dec}} \quad \text{for all } t \geq \tau_{\text{dec}}.$$

What it is: Gate G2. Operational pass condition for decoherence.

16. **Born-weight ansatz**

$$P(i \mid \psi) = F(x_i), \quad x_i \equiv |c_i|^2, \quad |\psi\rangle = \sum_i c_i |i\rangle.$$

What it is: Setup for the Born-rule functional-equation derivation.

17. **Compositionality functional equation**

$$F(xy) = F(x)F(y).$$

What it is: Product-state factorization constraint on the probability map.

18. **Normalization constraints on the weight function**

$$F(1) = 1, \quad F(0) = 0.$$

What it is: Boundary conditions for the Born-weight function.

19. **Additivity for exclusive two-outcome measurements**

$$F(x) + F(1 - x) = 1.$$

What it is: Exclusive-outcome normalization constraint.

20. **Multiplicative continuous solution family**

$$F(x) = x^\alpha, \quad \alpha \geq 0.$$

What it is: General continuous solution to the multiplicative functional equation.

21. **Born-rule uniqueness**

$$F(x) = x.$$

What it is: Unique solution after imposing the additivity constraint.

22. **Born rule**

$$P(i \mid \psi) = |c_i|^2.$$

What it is: Theorem: unique symmetry/compositionality-consistent probability assignment.

23. **Coarse-grained observable**

$$\tilde{A}(x) = \int_{\mathbb{R}} W_{\Delta x}(x - x') A(x') dx'.$$

What it is: Windowed observable measured at finite spatial resolution.

24. **Windowed interference factor**

$$\int_{\mathbb{R}} W_{\Delta x}(x) e^{i\Delta k x} dx = \widehat{W}_{\Delta x}(\Delta k).$$

What it is: Fourier-domain suppression factor for oscillatory interference terms.

25. **Top-hat window suppression law**

$$\widehat{W}_{\Delta x}(\Delta k) = \text{sinc}\left(\frac{\Delta k \Delta x}{2}\right).$$

What it is: Explicit suppression factor for a top-hat coarse-graining window.

26. **Finite-resolution classicality condition**

$$\Delta k \Delta x \gg 1.$$

What it is: Regime where oscillatory interference is strongly suppressed by coarse-graining.

27. **Measurement as a CPTP channel**

$$\mathcal{E}(\rho_S) = \text{Tr}_E[U(\rho_S \otimes \rho_E)U^\dagger].$$

What it is: General operational representation of measurement/coarse-graining as a channel.

28. **Ideal dephasing channel**

$$\mathcal{E}(\rho) = \sum_i \Pi_i \rho \Pi_i.$$

What it is: Ideal pointer-basis dephasing limit of the measurement channel.

29. **Observation-window coarse average**

$$\rho_M = \int W(\xi) \rho_J(\xi) d\mu(\xi).$$

What it is: Operational definition of bounded observation as a coarse average of fine-grained states.

30. **Born weights for Target M6**

$$p_i = \text{Tr}(\Pi_i \rho).$$

What it is: The target outcome weights against which frequency agreement is tested.

31. **Empirical frequencies for Target M6**

$$f_i = \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{i_n = i\}.$$

What it is: Empirical frequencies over N repeated trials.

32. **KL divergence meter**

$$\text{KL}(f\|p) = \sum_{i=1}^d f_i \log\left(\frac{f_i}{p_i}\right).$$

What it is: The Target M6 divergence metric in nats.

33. **Born-frequency gate**

$$\text{KL}(f\|p) \leq \varepsilon_{\text{KL}}.$$

What it is: Gate G3. Pass condition for the Born frequency meter.

34. **Expected KL scaling**

$$\mathbb{E}[\text{KL}(f\|p)] \approx \frac{d-1}{2N}.$$

What it is: Large- N asymptotic scaling estimate for the frequency meter.

35. **Sample-size planning rule**

$$N \gtrsim \frac{d}{2\varepsilon_{\text{KL}}}.$$

What it is: Conservative rule-of-thumb derived from the expected KL scaling.

36. **Method-of-types probability bound**

$$\Pr(\text{KL}(f\|p) \geq \varepsilon) \leq (N+1)^d e^{-N\varepsilon}.$$

What it is: High-probability bound controlling large deviations of empirical frequencies from Born weights.

37. **Robustness perturbation condition**

$$\frac{\|\delta H_{SE}\|}{\|H_{SE}\|} \leq \eta.$$

What it is: Declared perturbation size used in the robustness gate.

38. **Robustness gate**

small perturbations δH_{SE} induce only tolerance-bounded changes in Π_i and p_i .

What it is: Gate G4. Stability requirement for a mechanism to count as an instrument.

10 CF08 - Spinor Emergence via Domain-Wall Fermions

1. Ginsparg–Wilson gate

$$\|\{D, \gamma_5\} - aD\gamma_5D\|_\infty \leq 10^{-12}.$$

What it is: Gate P1. Exact/near-exact lattice chiral-symmetry protection requirement.

2. Residual-mass suppression gate

$$\frac{m_{\text{res}}(L_5)}{m_{\text{res}}(L_5/2)} \leq e^{-\lambda L_5/2}, \quad \lambda \geq 0.1/a.$$

What it is: Gate P2. Exponential chirality-leakage suppression with wall separation.

3. Dispersion-linearity gate

$$R^2 \geq 0.9999 \quad \text{for } |p| < \pi/(10a).$$

What it is: Gate P3. Low-momentum wall mode must be Dirac-like.

4. Rotational-isotropy gate

$$\frac{\Delta E}{E} \leq 10^{-3} \quad \text{at fixed } |p| = 0.1\pi/a.$$

What it is: Gate P4. Low-energy isotropy / Lorentz-compatibility condition.

5. Encoding-locality gate

$$\text{operator support} \leq C \log^2 N \quad \text{over } N \in \{32^3, 64^3, 128^3\}.$$

What it is: Gate P5. Fermionization/encoding must preserve locality up to polylogarithmic overhead.

6. VDM scalar lattice action

$$S_{\text{lattice}} = \sum_{n,i} \left[\frac{1}{2} \left(\frac{W_i^{n+1} - W_i^n}{\Delta t} \right)^2 - \frac{J}{2} \sum_{j \in N(i)} (W_j^n - W_i^n)^2 - V(W_i^n) \right] \Delta t.$$

What it is: Conservative scalar-lattice starting point for the matter-sector construction.

7. Continuum long-wavelength action

$$S_{\text{cont}} = \int dt d^3x \left[\frac{1}{2} (\partial_t \phi)^2 - \frac{c^2}{2} |\nabla \phi|^2 - V(\phi) \right], \quad c^2 = 2Ja^2.$$

What it is: Continuum limit of the scalar lattice action.

8. **Tachyonic quartic potential**

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4, \quad \mu^2 > 0, \lambda > 0.$$

What it is: Standard double-well/tachyonic potential admitting kink solutions.

9. **Vacua of the quartic potential**

$$\phi_{\pm} = \pm \frac{\mu}{\sqrt{\lambda}}.$$

What it is: Stable asymptotic vacua joined by the kink/domain wall.

10. **Static kink Euler–Lagrange equation**

$$-c^2 \frac{d^2 \phi_{\text{bg}}}{dz^2} + V'(\phi_{\text{bg}}) = 0.$$

What it is: Background equation defining the domain-wall profile.

11. **Kink solution**

$$\phi_{\text{bg}}(z) = \phi_+ \tanh\left(\frac{z - z_0}{\xi}\right), \quad \xi = \frac{c}{\mu}, \quad \phi_+ = \frac{\mu}{\sqrt{\lambda}}.$$

What it is: Standard continuous kink profile for the quartic potential.

12. **Exponential tail of the kink**

$$|\phi_{\text{bg}}(z) - \phi_+| \sim 2\phi_+ e^{-|z|/\xi} \quad (|z| \gg \xi).$$

What it is: Large- $|z|$ asymptotic approach of the kink to the vacuum.

13. **Wall decay rate**

$$\lambda \equiv \xi^{-1} = \mu/c.$$

What it is: Exponential localization rate governing tail decay and tunneling suppression.

14. **Discrete kink equation on the auxiliary lattice**

$$-c^2 \frac{\phi_{k+1} - 2\phi_k + \phi_{k-1}}{a_5^2} + V'(\phi_k) = 0.$$

What it is: Finite-difference form of the kink equation in the auxiliary z -direction.

15. **Domain-wall preflight gate**

$$R_{\text{tail fit}}^2 \geq 0.99, \quad \lambda \geq 0.1/a.$$

What it is: Gate G0. Kink must be visibly exponential and sufficiently localized before chirality gates are meaningful.

16. **Quadratic fluctuation action**

$$S_{\text{lin}} = \int dt d^3x dz \left[\frac{1}{2} (\partial_t \delta\phi)^2 - \frac{c^2}{2} (|\nabla \delta\phi|^2 + |\partial_z \delta\phi|^2) - \frac{1}{2} V''(\phi_{\text{bg}}) \delta\phi^2 \right].$$

What it is: Linearized action for fluctuations around the kink background.

17. **Bound-state equation in the auxiliary direction**

$$\left[-c^2 \frac{d^2}{dz^2} + V''(\phi_{\text{bg}}(z)) \right] \chi(z) = \epsilon \chi(z).$$

What it is: Schrödinger-type eigenvalue problem controlling wall-localized modes.

18. **Wall-parallel dispersion law**

$$E^2(\mathbf{p}) = \epsilon + c^2 |\mathbf{p}|^2.$$

What it is: Dispersion relation for a mode with auxiliary eigenvalue ϵ .

19. **Wall zero mode**

$$\chi_0(z) \propto \frac{d\phi_{\text{bg}}}{dz} \sim \text{sech}^2(z/\xi).$$

What it is: Normalizable wall-localized zero mode bound to the kink.

20. **Low-momentum linear dispersion**

$$E(\mathbf{p}) = v_F |\mathbf{p}| + \mathcal{O}(|\mathbf{p}|^3), \quad v_F = c + \mathcal{O}(a^2).$$

What it is: Theorem. Effective low-energy wall mode is Dirac-like.

21. **Ginsparg–Wilson relation**

$$\{D, \gamma_5\} = aD\gamma_5D.$$

What it is: Modified lattice chiral-symmetry condition satisfied by the overlap operator.

22. **Overlap operator**

$$D_{\text{ov}} = \frac{1}{a} \left(1 + \gamma_5 \text{sign}(H_W) \right).$$

What it is: Effective 3 + 1D operator induced from the 4 + 1D domain-wall construction.

23. **Wilson-Hermitian kernel**

$$H_W = \gamma_5 D_W.$$

What it is: Hermitian Wilson kernel used in the overlap construction.

24. **Equivalent overlap-operator form**

$$D_{\text{ov}} = \frac{1}{a} \left(1 - \frac{H_W}{\sqrt{H_W^\dagger H_W}} \right).$$

What it is: Polar-decomposition form of the overlap operator.

25. **Finite-wall Ginsparg–Wilson residual bound**

$$\|\{D, \gamma_5\} - aD\gamma_5D\|_\infty \leq C_1 a^2 + C_2 e^{-\lambda L_5}.$$

What it is: Quantitative finite- L_5 defect bound for approximate chiral protection.

26. **Wall-mode overlap tunneling bound**

$$\langle \chi_L, \chi_R \rangle \lesssim L_5 e^{-\lambda L_5}.$$

What it is: Theorem. Overlap of separated left/right wall modes is exponentially suppressed.

27. **Residual-mass scaling law**

$$m_{\text{res}} \sim A e^{-\lambda L_5}.$$

What it is: Theorem. Chirality leakage decays exponentially with wall separation.

28. **Bravyi–Kitaev support scaling**

$$|P(i)|, |U(i)| = \mathcal{O}(\log N), \quad \text{overall support} = \mathcal{O}(\log^2 N).$$

What it is: Operational locality scaling of the Bravyi–Kitaev transformation in the construction.

29. **Hypercubic lattice dispersion**

$$E^2(\mathbf{p}) \propto \sum_{\mu=1}^3 \sin^2(p_\mu a).$$

What it is: High-momentum lattice dispersion showing hypercubic anisotropy away from the low-energy regime.

11 CF09 - Gauge Field Emergence via Berry Connection

1. **Phase redundancy of the low-energy state bundle**

$$|\psi(x)\rangle \mapsto e^{i\Lambda(x)} |\psi(x)\rangle.$$

What it is: Local $U(1)$ phase redundancy of the effective low-energy states.

2. Quantum geometric tensor

$$Q_{\mu\nu}(x) \equiv \langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle.$$

What it is: The QGT used to extract the emergent gauge geometry.

3. Berry connection as emergent gauge potential

$$A_\mu(x) \equiv i \langle \psi(x) | \partial_\mu \psi(x) \rangle.$$

What it is: Definition of the emergent U(1) connection from the state bundle.

4. Berry-curvature / QGT relation

$$F_{\mu\nu}(x) \equiv \partial_\mu A_\nu - \partial_\nu A_\mu = 2 \text{Im} Q_{\mu\nu}(x).$$

What it is: Identification of the electromagnetic field strength with the Berry curvature.

5. Gauge transformation law for the Berry connection

$$A_\mu(x) \mapsto A_\mu(x) + \partial_\mu \Lambda(x).$$

What it is: The U(1) gauge transformation induced by local phase redefinition of the state.

6. Wilson-line phase

$$\mathcal{W}[\gamma] = \exp\left(-i \int_\gamma A_\mu dx^\mu\right).$$

What it is: Holonomy/parallel-transport phase generated by the emergent connection.

7. Lattice link variable from nearest-neighbor overlaps

$$U_{n,\mu} \equiv \frac{\langle \psi_n | \psi_{n+\hat{\mu}} \rangle}{|\langle \psi_n | \psi_{n+\hat{\mu}} \rangle|} \in \text{U}(1).$$

What it is: Lattice U(1) connection extracted from overlap phases.

8. Lattice gauge transformation law for the links

$$U_{n,\mu} \mapsto e^{i(\Lambda_{n+\hat{\mu}} - \Lambda_n)} U_{n,\mu}.$$

What it is: Standard lattice U(1) covariance of the overlap-defined links.

9. Lattice gauge potential from the link phase

$$A_{n,\mu} \equiv -\frac{1}{a} \text{Arg}(U_{n,\mu}), \quad U_{n,\mu} = e^{-iaA_{n,\mu}}.$$

What it is: Lattice gauge-potential representative associated with the overlap links.

10. **Plaquette variable**

$$U_{n,\mu\nu} \equiv U_{n,\mu} U_{n+\hat{\mu},\nu} U_{n+\hat{\nu},\mu}^* U_{n,\nu}^*.$$

What it is: The gauge-invariant oriented plaquette product.

11. **Discrete lattice curvature**

$$F_{n,\mu\nu} \equiv -\frac{1}{a^2} \text{Arg}(U_{n,\mu\nu}).$$

What it is: Lattice field-strength extracted from the plaquette phase.

12. **Continuum field-strength definition**

$$F_{\mu\nu}(x) = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

What it is: Continuum electromagnetic field-strength tensor.

13. **Berry-curvature inner-product form**

$$F_{\mu\nu} = i(\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\nu \psi | \partial_\mu \psi \rangle).$$

What it is: Explicit state-overlap expression for the curvature.

14. **Bianchi identity**

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.$$

What it is: Closure identity satisfied by the curvature two-form.

15. **Electric field from the curvature**

$$E^i = F^{0i}.$$

What it is: Identification of the electric-field components.

16. **Magnetic field from the curvature**

$$B^i = \frac{1}{2} \epsilon^{ijk} F_{jk}.$$

What it is: Identification of the magnetic-field components.

17. **Low-energy effective-action expansion**

$$S_{\text{eff}}[A] = \int d^4x \left[-\frac{1}{4g^2} F_{\mu\nu} F^{\mu\nu} + c_1 (\partial_\alpha F_{\mu\nu})(\partial^\alpha F^{\mu\nu}) + \dots \right].$$

What it is: Gauge-invariant derivative expansion whose leading operator is Maxwell F^2 .

18. **Emergent-coupling scaling hypothesis**

$$\frac{1}{g^2} \propto \frac{1}{a^2} \int dz |\chi_0(z)|^4.$$

What it is: Scaling estimate connecting the effective Maxwell coefficient to the localized zero-mode profile.

19. **Helmholtz decomposition of the gauge field**

$$\vec{A} = \vec{A}_\perp + \nabla\chi, \quad \nabla \cdot \vec{A}_\perp = 0.$$

What it is: Decomposition separating physical transverse content from pure-gauge longitudinal content.

20. **Pure-gauge implication**

$$A_\mu = \partial_\mu \phi \implies F_{\mu\nu} = 0.$$

What it is: The theorem used to argue that nontrivial curvature implies non-pure-gauge content.

21. **Coulomb potential**

$$\phi(\vec{x}) = \frac{q}{4\pi|\vec{x}|}, \quad V(r) \propto \frac{1}{r}.$$

What it is: Static long-range potential implied by a gapless Maxwell sector.

22. **Gauge-covariance / plaquette-invariance gate**

$$\text{mean plaquette phase residual under random gauge} \leq 10^{-10}.$$

What it is: Gate G1. Decisive test that plaquette curvature is gauge-invariant.

23. **Bianchi-residual gate**

$$\|\partial_{[\lambda} F_{\mu\nu]}\| \leq 10^{-10}.$$

What it is: Gate G2. Discrete or continuum Bianchi residual must vanish to tolerance.

24. **Maxwell-operator-dominance gate**

$$R^2 \geq 0.99$$

for a fit identifying a leading F^2 term on the declared IR window. *What it is:* Gate G3. The effective action must be dominated by the Maxwell operator.

25. **Transversality gate**

$$f_{\perp} \geq 0.999.$$

What it is: Gate G4. Physical gauge-field content must be overwhelmingly transverse.

26. **Gaplessness / photon-mass fit**

$$\omega^2(k) = c^2 k^2 + m_{\gamma}^2.$$

What it is: Dispersion model used to test whether the emergent photon is gapless.

27. **Photon-mass bound gate**

$$m_{\gamma} a < 10^{-12}.$$

What it is: Gate G5. The fitted photon gap must be below tolerance.

28. **Coulomb-vs-Yukawa gate**

$$\text{potential fit residual} \leq 1\% \quad \text{over } r \in [2a, 10a].$$

What it is: Gate G6. The static potential must favor Coulomb behavior over a Yukawa tail on the declared window.

12 CF10 - VDM Lattice Hydrodynamics, Continuum Limit, and Regularity Program

1. **Microscopic mass and momentum moments**

$$\rho(x, t_n) = \sum_i f_i(x, t_n), \quad \rho u_{\alpha}(x, t_n) = \sum_i c_{i,\alpha} f_i(x, t_n).$$

What it is: Definition of the macroscopic density and momentum from the lattice populations.

2. **Single-step lattice metriplectic update**

$$f_i^{n+1}(x + c_i \Delta t) = f_i^n(x) + \Delta t (J_i[f] + M_i[f]).$$

What it is: The lattice fluid update written as conservative plus dissipative limbs.

3. **BGK collision term**

$$M_i[f] = -\frac{1}{\tau} (f_i - f_i^{\text{eq}}(\rho, u)).$$

What it is: Single-relaxation-time dissipative limb.

4. **Collision conservation of mass and momentum**

$$\sum_i M_i[f] = 0, \quad \sum_i c_i M_i[f] = 0.$$

What it is: Built-in conserved moments of the collision operator.

5. **Discrete entropy / H-functional**

$$\Sigma[f] = - \sum_{x,i} f_i(x) \ln f_i(x).$$

What it is: Entropy-like functional monitored by the discrete gates.

6. **Positivity gate**

$$\min_{x,i} f_i(x, t_n) \geq -10^{-14}.$$

What it is: Gate F1.1 positivity threshold.

7. **Mass-drift gate**

$$\max_n \frac{|M_n - M_0|}{\max(\varepsilon, |M_0|)} \leq 10^{-12}, \quad M_n = \sum_x \rho(x, t_n).$$

What it is: Gate F1.1 mass-conservation threshold.

8. **Momentum-drift gate**

$$\max_n \frac{\|P_n - P_0\|}{\max(\varepsilon, \|P_0\|)} \leq 10^{-12}.$$

What it is: Gate F1.1 momentum-conservation threshold where applicable.

9. **Per-step entropy-production gate**

$$\Delta \Sigma_n \geq -10^{-12}.$$

What it is: Gate F1.1 collision-segment entropy monotonicity threshold.

10. **Degeneracy-residual gates**

$$\|J\nabla\Sigma\|_\infty \leq 10^{-12}, \quad \|M\nabla I\|_\infty \leq 10^{-12}.$$

What it is: Gate F1.1 metriplectic identity checks.

11. **Chapman–Enskog expansion**

$$f_i = f_i^{(0)} + \epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \dots$$

What it is: Multiscale expansion used to derive the continuum hydrodynamic limit.

12. **Viscosity relation**

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right) \Delta t.$$

What it is: Effective kinematic viscosity of the lattice scheme in the hydrodynamic limit.

13. **Incompressible Navier–Stokes target**

$$\partial_t u + (u \cdot \nabla) u = -\frac{1}{\rho_0} \nabla p + \nu \Delta u.$$

What it is: Continuum target PDE recovered in the low-Mach / low-Knudsen regime.

14. **Taylor–Green viscosity-recovery gate**

$$\text{rel_err}_\nu \leq 0.05.$$

What it is: Gate F1.2 threshold for viscosity recovery from Taylor–Green decay.

15. **Lid-cavity incompressibility gate**

$$\max_t \|\nabla \cdot u\|_2 \leq 10^{-6}.$$

What it is: Gate F1.2 divergence threshold for the lid-driven cavity benchmark.

16. **Littlewood–Paley decomposition**

$$u = \sum_{k \in \mathbb{Z}} u_k, \quad u_k = \mathcal{P}_k u.$$

What it is: Dyadic shell decomposition used in the regularity program.

17. **Shell vorticity**

$$\omega_k = \nabla \times u_k.$$

What it is: Dyadic vorticity component on shell k .

18. **Shell enstrophy**

$$\Omega_k(t) = \int |\omega_k(x, t)|^2 dx = \|\omega_k(\cdot, t)\|_2^2.$$

What it is: Dyadic-shell enstrophy observable.

19. **Navier–Stokes energy balance**

$$\frac{dE}{dt} = -\nu \|\nabla u\|_2^2.$$

What it is: Global viscous energy dissipation law.

20. **Scale-wise dissipation cost**

$$\epsilon_k(t) \sim \nu 2^{2k} \Omega_k(t).$$

What it is: Dyadic-shell dissipation weight.

21. **Vorticity equation**

$$\partial_t \omega = \nabla \times (u \times \omega) + \nu \Delta \omega.$$

What it is: Vorticity evolution form of incompressible Navier–Stokes.

22. **Shell enstrophy balance heuristic**

$$\frac{d\Omega_k}{dt} = \mathcal{T}_k[u] - \nu 2^{2k} \Omega_k + (\text{commutator terms}).$$

What it is: Scale-local transfer-versus-dissipation balance.

23. **Time-uniform shell tail decay conjecture**

$$\Omega_k(t) \leq C 2^{-\beta k} \quad \text{for all } t \geq 0, \ k \geq k_0, \quad \beta > 3.$$

What it is: Conjecture F1.A, part (1). The A8-style time-uniform dyadic tail bound.

24. **Scale-wise transfer bound conjecture**

$$\mathcal{T}_k[u](t) \leq c_2 \nu 2^{2k} \Omega_k(t) \quad (k \geq k_0).$$

What it is: Conjecture F1.A, part (2). Dissipation dominates transfer at high shells.

25. **Tail-fit regression model**

$$\log_2 \Omega_k \approx a - \beta k.$$

What it is: Fitted log-linear tail law used in the spectrum gate.

26. **Tail-fit quality gate**

$$R^2 \geq 0.98.$$

What it is: Gate F1.3 quality threshold for the shell-tail fit.

27. **Criticality-margin gate**

$$\beta_{\min} \geq 3.2.$$

What it is: Gate F1.3 threshold placing the fitted tail exponent safely above the analytic $\beta > 3$ requirement.

28. **Conditional regularity theorem: finite enstrophy**

$$\Omega(t) = \|\omega(\cdot, t)\|_2^2 < \infty \quad \text{for all } t \geq 0.$$

What it is: First conclusion of the conditional regularity theorem under Conjecture F1.A.

29. **Conditional regularity theorem: integrable vorticity supremum norm**

$$\int_0^T \|\omega(\cdot, t)\|_\infty dt < \infty.$$

What it is: Second conclusion of the conditional regularity theorem.

30. **Tail summability condition for bounded enstrophy**

$$\sum_{k \geq k_0} C 2^{-\beta k} < \infty \quad (\beta > 1).$$

What it is: Lemma R1. Tail decay with $\beta > 1$ implies finite total enstrophy.

31. **Bernstein shell bound**

$$\|\omega_k\|_\infty \lesssim 2^{3k/2} \|\omega_k\|_2 = 2^{3k/2} \Omega_k^{1/2}.$$

What it is: Lemma R2. Dyadic-shell L^∞ control from L^2 control in 3D.

32. **Tail summability condition for bounded vorticity supremum norm**

$$\sum_{k \geq k_0} 2^{3k/2} 2^{-\beta k/2} < \infty \quad (\beta > 3).$$

What it is: Lemma R2. Tail decay with $\beta > 3$ yields a uniform $\|\omega\|_\infty$ bound.

33. **Beale–Kato–Majda criterion**

$$\int_0^T \|\omega(\cdot, t)\|_\infty dt < \infty \implies \text{no finite-time blow-up on } [0, T].$$

What it is: Lemma R3 / classical criterion used in the conditional theorem.

13 CF11 - Dark Sector Emergence from Metriplectic Dynamics

1. **Imported metriplectic evolution law**

$$\partial_t q = J(q) \frac{\delta \mathcal{I}}{\delta q} + M(q) \frac{\delta \Sigma}{\delta q}, \quad J^\top = -J, \quad M^\top = M \succeq 0.$$

What it is: The axiom-core evolution law imported from earlier CFs.

2. **Degeneracy conditions**

$$J(q) \frac{\delta \Sigma}{\delta q} = 0, \quad M(q) \frac{\delta \mathcal{I}}{\delta q} = 0.$$

What it is: The metriplectic degeneracy constraints used throughout CF11.

3. Entropy production law

$$\frac{d\Sigma}{dt} = \left\langle \frac{\delta\Sigma}{\delta q}, M(q) \frac{\delta\Sigma}{\delta q} \right\rangle \geq 0.$$

What it is: Imported H-theorem / entropy-law contract.

4. Hierarchy-throttled effective speed

$$c_{\text{eff}}(k) = c_0 e^{-\beta D_{\text{void}}(k)/2}, \quad D_{\text{void}}(k) \sim k.$$

What it is: Imported CF03 derived-limit speed-throttling law.

5. Telegraph cone speed

$$c = \sqrt{D/\tau}.$$

What it is: Imported CF04 finite-speed transport relation.

6. Proxy-model state

$$q = (\Phi, \Pi, u).$$

What it is: Minimal proxy state used for the dark-sector bookkeeping model.

7. Conserved functional in the proxy model

$$\mathcal{I}[\Phi, \Pi, u] = \int d^3x \left(\frac{1}{2} \Pi^2 + \frac{1}{2} |\nabla \Phi|^2 + V(\Phi) + u \right).$$

What it is: Conserved energy-like functional for the proxy model.

8. Entropy functional in the proxy model

$$\Sigma[\Phi, \Pi, u] = \int d^3x s(u), \quad s'(u) > 0.$$

What it is: Entropy functional for the proxy model.

9. Proxy J -operator

$$J = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

What it is: Canonical reversible limb acting on (Φ, Π) while leaving u untouched.

10. Proxy M -operator

$$M(q) = \frac{1}{\tau s'(u)} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -\Pi \\ 0 & -\Pi & \Pi^2 \end{pmatrix} = \frac{1}{\tau s'(u)} v v^\top, \quad v = (0, 1, -\Pi).$$

What it is: Rank-1 PSD dissipative limb satisfying the degeneracy constraint.

11. **Proxy-model entropy production**

$$\dot{\Sigma} = \left\langle \frac{\delta \Sigma}{\delta q}, M \frac{\delta \Sigma}{\delta q} \right\rangle = \int \frac{\Pi^2}{\tau} d^3x \geq 0.$$

What it is: Explicit entropy-growth law in the proxy model.

12. **Proxy evolution for the field**

$$\dot{\Phi} = \Pi.$$

What it is: First proxy evolution equation.

13. **Proxy evolution for the momentum**

$$\dot{\Pi} = \nabla^2 \Phi - V'(\Phi) - \frac{1}{\tau} \Pi.$$

What it is: Second proxy evolution equation.

14. **Proxy evolution for the reservoir energy**

$$\dot{u} = \frac{1}{\tau} \Pi^2.$$

What it is: Third proxy evolution equation.

15. **Reduced damped field equation**

$$\ddot{\Phi} + \frac{1}{\tau} \dot{\Phi} - \nabla^2 \Phi + V'(\Phi) = 0.$$

What it is: Reduced damped Klein–Gordon / Allen–Cahn type equation obtained by eliminating Π .

16. **Generator split**

$$\dot{q} = \dot{q}_J + \dot{q}_M, \quad \dot{q}_J := J \frac{\delta \mathcal{I}}{\delta q}, \quad \dot{q}_M := M \frac{\delta \Sigma}{\delta q}.$$

What it is: Split of motion into reversible and irreversible generator limbs.

17. **Linearized perturbation equation**

$$\partial_t(\delta q) = \mathcal{L}(t) \delta q + \mathcal{N}(\delta q).$$

What it is: Linearization around a coarse-grained background $\bar{q}(t)$.

18. **State-level mode split via spectral projectors**

$$\delta q = \delta q_J + \delta q_M, \quad \delta q_J := P_J \delta q, \quad \delta q_M := P_M \delta q, \quad P_J P_M = 0.$$

What it is: The mathematically standard state split used in CF11.

19. **Quadratic perturbation energy**

$$\delta^2 \mathcal{I}[\delta q] = \frac{1}{2} \langle \delta q, H_{\mathcal{I}}(\bar{q}) \delta q \rangle.$$

What it is: Hessian-level quadratic energy around the background state.

20. **Sector energies**

$$E_J = \frac{1}{2} \langle \delta q_J, H_{\mathcal{I}} \delta q_J \rangle, \quad E_M = \frac{1}{2} \langle \delta q_M, H_{\mathcal{I}} \delta q_M \rangle.$$

What it is: Quadratic energies assigned to the J -sector and M -sector.

21. **Quadratic interaction remainder**

$$E_{\times} = \langle \delta q_J, H_{\mathcal{I}} \delta q_M \rangle.$$

What it is: Cross-term energy remainder between the projected sectors.

22. **Scalar-field proxy energy density**

$$\rho[\Phi] = \frac{1}{2} \dot{\Phi}^2 + \frac{1}{2a^2} |\nabla \Phi|^2 + V(\Phi).$$

What it is: Effective energy-density diagnostic used in the derived-limit cosmology bookkeeping.

23. **Scalar-field proxy pressure**

$$p[\Phi] = \frac{1}{2} \dot{\Phi}^2 - \frac{1}{6a^2} |\nabla \Phi|^2 - V(\Phi).$$

What it is: Effective pressure diagnostic used in the derived-limit cosmology bookkeeping.

24. **Sector densities and pressures**

$$\rho_J := \langle \rho[\Phi_J] \rangle_{\ell}, \quad p_J := \langle p[\Phi_J] \rangle_{\ell},$$

$$\rho_M := \langle \rho[\Phi_M] \rangle_{\ell}, \quad p_M := \langle p[\Phi_M] \rangle_{\ell}.$$

What it is: Coarse-grained J -sector and M -sector effective densities and pressures.

25. **Tracked density and pressure remainders**

$$\rho_{\times} := \langle \rho[\Phi] - \rho[\Phi_J] - \rho[\Phi_M] \rangle_{\ell},$$

$$p_{\times} := \langle p[\Phi] - p[\Phi_J] - p[\Phi_M] \rangle_{\ell}.$$

What it is: Explicitly tracked interaction remainders.

26. ***J*-sector kinetic, gradient, and potential diagnostics**

$$K_J := \frac{1}{2} \langle \dot{\Phi}_J^2 \rangle_\ell, \quad G_J := \frac{1}{2a^2} \langle |\nabla \Phi_J|^2 \rangle_\ell, \quad U_J := \langle V(\Phi_J) \rangle_\ell.$$

What it is: Diagnostic decomposition used to control the $w_J \approx -1$ regime.

27. ***J*-sector density and pressure from diagnostics**

$$\rho_J = K_J + G_J + U_J, \quad p_J = K_J - \frac{1}{3} G_J - U_J.$$

What it is: Exact sector expressions entering the *J*-sector EoS.

28. ***J*-sector small parameters**

$$\epsilon_J := \frac{K_J}{U_J}, \quad \delta_J := \frac{G_J}{U_J}.$$

What it is: Dimensionless regime-diagnostic ratios for the vacuum-like limit.

29. ***J*-sector exact EoS identity**

$$w_J = \frac{\epsilon_J - \frac{1}{3} \delta_J - 1}{\epsilon_J + \delta_J + 1}.$$

What it is: Exact equation-of-state identity for the *J*-sector.

30. ***J*-sector EoS bound**

$$|1 + w_J| \leq \frac{2\epsilon_{\max} + \frac{2}{3}\delta_{\max}}{1 - \epsilon_{\max} - \delta_{\max}}.$$

What it is: Bound showing that $w_J \approx -1$ is a measured consequence only when ϵ_J, δ_J are small.

31. ***M*-sector defect-gas dust limit**

$$w_M \approx \frac{\sigma_v^2}{3}.$$

What it is: Dust-like EoS in the nonrelativistic defect/particle regime.

32. **Quadratic-minimum approximation**

$$V(\Phi) \approx \frac{1}{2} m^2 (\Phi - \Phi_\star)^2.$$

What it is: Local quadratic approximation near a stable minimum.

33. **Rapid-oscillation dust limit**

$$\langle K_M \rangle \approx \langle U_M \rangle \implies \langle w_M \rangle \approx 0.$$

What it is: Dust-like EoS in the rapid-oscillation regime near a quadratic minimum.

34. **Exchange current definition**

$$Q^\nu := -\nabla_\mu T_J^{\mu\nu}.$$

What it is: Internal bookkeeping definition of the exchange current after the sector split.

35. **Total-stress bookkeeping identity**

$$\nabla_\mu (T_J^{\mu\nu} + T_M^{\mu\nu}) = -\nabla_\mu T_\times^{\mu\nu}.$$

What it is: Exact conservation bookkeeping including the tracked remainder.

36. **Effective interacting-fluid limit**

$$\nabla_\mu T_J^{\mu\nu} \approx -Q^\nu, \quad \nabla_\mu T_M^{\mu\nu} \approx +Q^\nu.$$

What it is: Effective two-fluid description when $T_\times^{\mu\nu}$ is small by gate.

37. **Hierarchy–lensing bias statistic**

$$\beta_{\text{bias}} := \frac{\kappa_{\text{obs}} - \kappa_{\text{baseline}}}{\kappa_{\text{baseline}}}.$$

What it is: Minimal observational statistic linking hierarchy indicators to lensing bias.

38. **Degeneracy-residual gate**

$$\frac{\|J \delta \Sigma\|}{\|\delta \Sigma\|} \leq 10^{-12}, \quad \frac{\|M \delta \mathcal{I}\|}{\|\delta \mathcal{I}\|} \leq 10^{-12}.$$

What it is: T2 gate G-CF11-T2-1 for metriplectic instrument validity.

39. **Entropy-law gate**

$$\Delta \Sigma \geq -\varepsilon_\Sigma.$$

What it is: T2 gate G-CF11-T2-2 for discrete entropy monotonicity.

40. **Spectral-gap gate**

$$\Re \lambda \leq -\gamma$$

for all declared relaxational modes. *What it is:* T2 gate G-CF11-T2-3 ensuring the mode split is meaningful.

41. **Cross-term control gate**

$$\frac{|\rho_\times|}{\rho_J + \rho_M} \leq \varepsilon_\times, \quad \varepsilon_\times = 0.05 \text{ (default)}.$$

What it is: T2 gate G-CF11-T2-4 constraining uncontrolled interaction remainders.

42. ***J*-sector regime-discipline gate**

$$\epsilon_J, \delta_J \leq 0.05.$$

What it is: Default T2 gate G-CF11-T2-5 condition before applying the $w_J \approx -1$ band test.

43. **Causality / cone-bound gate**

$$v_{\text{front}} \leq c_{\text{eff}}(1 + \eta), \quad \eta = 0.02 \text{ (default)}.$$

What it is: T2 gate G-CF11-T2-6 for finite-speed propagation.

44. **Hierarchy–lensing detection/null gate**

$$|\beta_{\text{bias}}| > 5\sigma \quad \text{or} \quad |\beta_{\text{bias}}| < 1\sigma.$$

What it is: T3+ gate G-CF11-T3-1 for the hierarchy–lensing channel.

45. **Model-selection gate**

$$\Delta\text{AIC} \leq -2 \quad \text{or} \quad \Delta\text{AIC} \geq +10.$$

What it is: Typical interpretation thresholds stated for T3+ gate G-CF11-T3-2.

46. **Interacting-fluid closure gate**

$$\theta > 3\sigma \quad \text{with} \quad \dot{\Sigma} \geq 0 \quad \text{or} \quad |\theta| < 1\sigma.$$

What it is: T3+ gate G-CF11-T3-3 for a pre-registered effective coupling $Q(z; \theta)$.

47. **Jacobian / linearized-generator decomposition step**

$$L_i = QTQ^\top.$$

What it is: Real Schur decomposition used in the lattice implementation of the spectral split.

48. **Canon-entry mode split**

$$\delta q = P_J \delta q + P_M \delta q, \quad P_J P_M = 0.$$

What it is: Proposed canon equation VDM-E-145.

49. **Canon-entry sector energies**

$$E_J = \frac{1}{2} \langle \delta q_J, H_{\mathcal{I}} \delta q_J \rangle, \quad E_M = \frac{1}{2} \langle \delta q_M, H_{\mathcal{I}} \delta q_M \rangle.$$

What it is: Proposed canon equation VDM-E-146.

50. **Canon-entry J -sector EoS formulas**

$$w_J = \frac{\epsilon_J - \frac{1}{3}\delta_J - 1}{\epsilon_J + \delta_J + 1}, \quad |1 + w_J| \leq \frac{2\epsilon_{\max} + \frac{2}{3}\delta_{\max}}{1 - \epsilon_{\max} - \delta_{\max}}.$$

What it is: Proposed canon equation VDM-E-147.

51. **Canon-entry dust limits**

$$w_M \approx \frac{\sigma_v^2}{3}, \quad \langle w_M \rangle \approx 0.$$

What it is: Proposed canon equation VDM-E-148.

52. **Canon-entry exchange current**

$$Q^\nu := -\nabla_\mu T_J^{\mu\nu}.$$

What it is: Proposed canon equation VDM-E-149.

14 CF12 - Gravity, Black Holes, and Saturation Geometry

1. **Invariant burden on a carrier region**

$$\mathcal{B}[\Omega] := \int_{\Omega} \rho_{\text{inv}} dV_g.$$

What it is: Definition of the total invariant-bearing burden hosted in a compact carrier region.

2. **Inertial and gravitational mass functionals**

$$m_I(\Omega) := c^{-2}\mathcal{R}_J[\Omega], \quad m_G(\Omega) := c^{-2}\mathcal{R}_M[\Omega].$$

What it is: Operational mass functionals defined from the reversible and irreversible readings of the same burden host.

3. **Equivalence principle from single-burden identity**

$$m_I(\Omega) = m_G(\Omega) = c^{-2}\mathcal{B}[\Omega].$$

What it is: Theorem of CF12: inertial and gravitational mass are two measurements of the same burden functional.

4. **Local Lorentzian cone metric**

$$ds^2 = -c_*^2 d\tau^2 + h_{ij}(x) dx^i dx^j.$$

What it is: Minimal quadratic encoding of the inherited finite-speed transport cone.

5. **Null-cone / eikonal relation**

$$g^{\mu\nu} \partial_\mu S \partial_\nu S = 0.$$

What it is: Dual statement that propagation fronts are null with respect to the derived Lorentzian metric.

6. **Carrier four-velocity normalization**

$$g_{\mu\nu} u^\mu u^\nu = -c_*^2.$$

What it is: Normalization of the future-directed carrier four-velocity field.

7. **Spatial projector orthogonal to the carrier flow**

$$P_{\mu\nu} := g_{\mu\nu} + c_*^{-2} u_\mu u_\nu.$$

What it is: Spatial projector orthogonal to u^μ .

8. **Invariant source tensor**

$$T_{\mu\nu} = \rho_{\text{inv}} u_\mu u_\nu + p_{\text{inv}} P_{\mu\nu} + \Pi_{\mu\nu} + q_{(\mu} u_{\nu)}.$$

What it is: Minimal symmetric source bookkeeping for storage, pressure, anisotropy, and flux of invariant burden.

9. **Saturation correction tensor**

$$\Delta_{\mu\nu} := \alpha (\nabla_\mu \nabla_\nu s - g_{\mu\nu} \square s) + \beta (\nabla_\mu s \nabla_\nu s - \tfrac{1}{2} g_{\mu\nu} (\nabla s)^2).$$

What it is: Explicit saturation-gradient correction sector added to Einstein-form closure.

10. **Effective gravitational field equation**

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \Delta_{\mu\nu} = \kappa_{\text{inv}} T_{\mu\nu}, \quad \kappa_{\text{inv}} = \frac{8\pi G_{\text{eff}}}{c^4}.$$

What it is: Completed CF12 gravitational field equation.

11. **Weak-field metric ansatz**

$$g_{00} = -c^2 \left(1 + \frac{2\Phi}{c^2} \right), \quad g_{ij} = \delta_{ij} \left(1 - \frac{2\Phi}{c^2} \right).$$

What it is: Quasistatic weak-field ansatz used to recover Newtonian gravity.

12. **Weak-field Poisson closure**

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{inv}}.$$

What it is: Newtonian-limit closure showing invariant burden density sources the gravitational potential.

13. **Radial saturation diagnostic**

$$\chi(r) := \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2r}.$$

What it is: Same-domain saturation ratio for a static spherical exterior channel.

14. **Static spherical metric ansatz**

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + e^{2\Lambda(r)}dr^2 + r^2d\Omega^2.$$

What it is: Standard static spherical ansatz used in the vacuum exterior derivation.

15. **Vacuum exterior radial factor**

$$e^{-2\Lambda(r)} = 1 - \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2r}.$$

What it is: Exterior vacuum solution for the radial metric factor.

16. **Vacuum exterior temporal factor**

$$e^{2\Phi(r)} = 1 - \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2r}.$$

What it is: Exterior vacuum solution for the time-redshift factor.

17. **Schwarzschild-form exterior metric**

$$ds^2 = -\left(1 - \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2r}\right)c^2dt^2 + \left(1 - \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

What it is: Completed static spherical exterior geometry in CF12.

18. **Horizon radius**

$$r_H = \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2}.$$

What it is: Horizon / saturation radius of the static spherical sector.

19. **Horizon saturation condition**

$$\chi(r_H) = 1.$$

What it is: The horizon is exactly the same-domain saturation threshold.

20. **Horizon area**

$$A_H = 4\pi r_H^2.$$

What it is: Area of the saturation boundary.

21. **Boundary articulation count**

$$N_H = \frac{A_H}{\ell_*^2}.$$

What it is: Number of fundamental articulation cells on the horizon boundary.

22. **Generic boundary-count entropy law**

$$S_{BH} = \eta k_B N_H = \eta k_B \frac{A_H}{\ell_*^2}.$$

What it is: Entropy as boundary articulation count before fixing the coefficient.

23. **Horizon area in terms of invariant mass**

$$A_H = \frac{16\pi G_{\text{eff}}^2 M_{\text{inv}}^2}{c^4}.$$

What it is: Mass-area relation for the static spherical horizon.

24. **Entropy derivative with respect to invariant mass**

$$\frac{dS_{BH}}{dM_{\text{inv}}} = 32\pi\eta k_B \frac{G_{\text{eff}}^2 M_{\text{inv}}}{c^4 \ell_*^2}.$$

What it is: Intermediate relation used to fix the entropy coefficient from the first law.

25. **Exterior lapse function**

$$f(r) := 1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r}, \quad g_{tt} = -c^2 f(r).$$

What it is: Lapse function used to compute the surface gravity.

26. **Surface gravity definition**

$$\kappa_H = \frac{c^2}{2} |f'(r_H)| = \frac{1}{2} |\partial_r g_{tt}|_{r_H}.$$

What it is: Horizon surface gravity in the static exterior geometry.

27. **Derivative of the lapse function**

$$f'(r) = \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r^2}.$$

What it is: Radial derivative used in the surface-gravity evaluation.

28. **Surface gravity law**

$$\kappa_H = \frac{c^4}{4G_{\text{eff}}M_{\text{inv}}}.$$

What it is: Completed surface-gravity formula for the static spherical horizon.

29. **Hawking temperature**

$$T_H = \frac{\hbar\kappa_H}{2\pi k_B c} = \frac{\hbar c^3}{8\pi k_B G_{\text{eff}}M_{\text{inv}}}.$$

What it is: Hawking temperature interpreted as the rate scale of forced orthogonal re-articulation at saturation.

30. **Fundamental gravitational articulation length**

$$\ell_*^2 := \frac{\hbar G_{\text{eff}}}{c^3}.$$

What it is: Natural gravitational articulation scale fixed by first-law consistency.

31. **Boundary-count coefficient**

$$\eta = \frac{1}{4}.$$

What it is: Coefficient fixed by matching the first law with the Hawking temperature.

32. **Completed Bekenstein–Hawking area law**

$$S_{BH} = \frac{k_B A_H}{4\ell_*^2} = \frac{k_B c^3 A_H}{4\hbar G_{\text{eff}}}.$$

What it is: Final black-hole entropy law of CF12.

33. **Black-hole first law**

$$dE = c^2 dM_{\text{inv}} = T_H dS_{BH}.$$

What it is: Thermodynamic first-law closure for the saturation boundary.

34. **Quasi-static evaporation law**

$$\frac{dM_{\text{inv}}}{dt} = -\frac{\gamma \hbar c^4}{G_{\text{eff}}^2 M_{\text{inv}}^2}.$$

What it is: Leading mass-loss law with greybody-corrected emission coefficient γ .

35. **Total articulation-record ledger**

$$\mathcal{I}_{\text{tot}} = \mathcal{I}_H + \mathcal{I}_R + \mathcal{I}_E.$$

What it is: Total information/articulation record split across horizon, radiation, and environment channels.

36. **Information-preservation ledger under evaporation**

$$\frac{d\mathcal{I}_{\text{tot}}}{dt} = 0, \quad \frac{d\mathcal{I}_H}{dt} < 0, \quad \frac{d\mathcal{I}_R}{dt} > 0.$$

What it is: Information-transfer law under horizon evaporation.

37. **Radiation-horizon transfer balance**

$$\frac{d\mathcal{I}_R}{dt} = -\frac{d\mathcal{I}_H}{dt}.$$

What it is: Transfer identity after subtracting any asymptotic leakage into the environment channel.

38. **Cumulative emitted coarse-grained entropy**

$$Q(t) := \int_0^t \frac{1}{T_H(t')} \left(-\frac{dE}{dt'} \right) dt' = \int_0^t \dot{S}_{\text{emit}}(t') dt'.$$

What it is: Emitted entropy ledger used to define the Page curve.

39. **Page-curve law**

$$S_R(t) = \min\{Q(t), S_{BH}(0) - Q(t)\}.$$

What it is: Completed Page-curve law of CF12.

40. **Page-time condition**

$$Q(t_P) = \frac{1}{2} S_{BH}(0).$$

What it is: Definition of the Page time.

41. **Cosmogenesis residual-closure law**

$$H^2(a) = H_{11}^2(a) + \xi \langle |\nabla \Phi|^2 \rangle_a + \zeta \langle \sigma_{\text{sat}} \rangle_a.$$

What it is: Explicit gravitational back-reaction correction to the CF11 cosmological background.

42. **Completed-formalism compact source tensor**

$$T_{\mu\nu} = \rho_{\text{inv}} u_\mu u_\nu + p_{\text{inv}} P_{\mu\nu} + \Pi_{\mu\nu} + q_{(\mu} u_{\nu)}.$$

What it is: Compact restatement of the carrier source bookkeeping in the completed-formalism summary.

43. **Completed-formalism equivalence principle**

$$m_I = m_G = c^{-2}\mathcal{B}.$$

What it is: Compact restatement of the equivalence principle in the completed-formalism summary.

44. **Completed-formalism Lorentzian cone metric**

$$ds^2 = -c_*^2 d\tau^2 + h_{ij} dx^i dx^j, \quad g^{\mu\nu} \partial_\mu S \partial_\nu S = 0.$$

What it is: Compact restatement of the causal metric and null-front condition.

45. **Completed-formalism field equation**

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \Delta_{\mu\nu} = \kappa_{\text{inv}} T_{\mu\nu}.$$

What it is: Compact restatement of the effective gravitational field equation.

46. **Completed-formalism weak-field closure**

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{inv}}.$$

What it is: Compact restatement of the Newtonian-limit closure.

47. **Completed-formalism static spherical exterior**

$$ds^2 = - \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right)^{-1} dr^2 + r^2 d\Omega^2.$$

What it is: Compact restatement of the exterior vacuum geometry.

48. **Completed-formalism horizon and area law**

$$r_H = \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2}, \quad S_{BH} = \frac{k_B A_H}{4\ell_*^2}, \quad \ell_*^2 = \frac{\hbar G_{\text{eff}}}{c^3}.$$

What it is: Compact restatement of the horizon radius and entropy law.

49. **Completed-formalism Hawking temperature and first law**

$$T_H = \frac{\hbar c^3}{8\pi k_B G_{\text{eff}} M_{\text{inv}}}, \quad d(M_{\text{inv}} c^2) = T_H dS_{BH}.$$

What it is: Compact restatement of the thermodynamic closure.

50. **Completed-formalism Page curve**

$$S_R(t) = \min\{Q(t), S_{BH}(0) - Q(t)\}, \quad Q(t) = \int_0^t \dot{S}_{\text{emit}} dt'.$$

What it is: Compact restatement of the information-transfer/Page-curve law.

51. **Completed-formalism cosmological residual closure**

$$H^2(a) = H_{11}^2(a) + \xi \langle |\nabla \Phi|^2 \rangle_a + \zeta \langle \sigma_{\text{sat}} \rangle_a.$$

What it is: Compact restatement of the CF11 residual-closure ansatz.

15 CF13 - π , Transcendence, Chirality, and the Nielsen–Ninomiya Obstruction

1. Branch spine

primitive invariant $\longrightarrow$ ORS quarter-turn $\longrightarrow \pi \longrightarrow$ half-turn irreconcilability $\longrightarrow$ chirality $\longrightarrow$ Nielsen–

What it is: The branch CF13 closes as one invariant-geometric chain.

2. Inherited ORS algebra seed

$$R^2 = -I.$$

What it is: The quarter-turn seed inherited from CF000/CF00 and used throughout CF13.

3. Continuous completion of ORS

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta).$$

What it is: Theorem 3.1. Continuous completion of the discrete ORS quarter-turn on the two-axis carrier.

4. Identity element of the continuous ORS family

$$R_0 = I.$$

What it is: Identity element of the continuous completion.

5. Quarter-turn recovery

$$R_{\pi/2} = R.$$

What it is: The continuous completion recovers the discrete quarter-turn seed.

6. Composition law of the continuous ORS family

$$R_{\theta+\phi} = R_\theta R_\phi.$$

What it is: Group law of the continuous rotational completion.

7. Primitive half-turn constant

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

What it is: Definition 3.2 / Theorem 3.2. π is the least positive rotation at which the ORS orbit reaches the opposite pole.

8. Least positive opposite-pole attainment set

$$\{\theta > 0 : R_\theta = -I\}.$$

What it is: The set whose unique least positive element defines the primitive half-turn constant.

9. **Full ORS orbit of a carrier point**

$$\mathcal{O}(p) := \{R_\theta p : \theta \in [0, 2\pi)\}.$$

What it is: Full closed orbit of a carrier point under the continuous ORS family.

10. **Area realization of the ORS half-turn constant**

$$\pi = \iint_{x^2+y^2 \leq 1} dx \, dy.$$

What it is: Corollary 3.2. Downstream geometric realization of π as the area of the unit ORS disk.

11. **Complex-action form of the ORS family**

$$R_\theta(z) = e^{i\theta} z.$$

What it is: Theorem 3.3. Complex-phase action of the continuous ORS family on the completed complex carrier.

12. **Half-turn closure identity**

$$e^{i\pi} = -1.$$

What it is: Theorem 3.3. Complex-expression of the exact half-turn to the opposite pole.

13. **Generic algebraic-function obstruction form**

$$a_n(\theta)f^n + a_{n-1}(\theta)f^{n-1} + \cdots + a_0(\theta) = 0, \quad a_i(\theta) \in \mathbb{Q}[\theta].$$

What it is: The type of polynomial equation Theorem A excludes for the ORS coordinate functions.

14. **ORS coordinate differential equation**

$$f'' + f = 0.$$

What it is: The ODE satisfied by the coordinate functions of the continuous ORS orbit.

15. **Initial data for the cosine branch**

$$\cos 0 = 1, \quad \cos'(0) = 0.$$

What it is: Initial data fixing the cosine projection of the ORS orbit.

16. **Initial data for the sine branch**

$$\sin 0 = 0, \quad \sin'(0) = 1.$$

What it is: Initial data fixing the sine projection of the ORS orbit.

17. **Theorem A coordinate-function parameterization**

$$\Gamma(\theta) = (\cos \theta, \sin \theta), \quad \theta \in [0, 2\pi).$$

What it is: Parameterization of the closed ORS orbit used in the transcendence branch.

18. **Area of the ORS orbit by line integral**

$$A = \frac{1}{2} \oint_{\Gamma} (x dy - y dx).$$

What it is: Theorem B. Standard area formula applied to the ORS orbit.

19. **Area reduction along the ORS orbit**

$$A = \frac{1}{2} \int_0^{2\pi} (\cos \theta \cdot \cos \theta - \sin \theta \cdot (-\sin \theta)) d\theta.$$

What it is: The explicit substitution step in Theorem B.

20. **Area simplification of the ORS orbit**

$$A = \frac{1}{2} \int_0^{2\pi} d\theta = \pi.$$

What it is: Theorem B. The enclosed area of the unit ORS orbit equals π .

21. **Transcendence claim for π**

$$\pi \notin \overline{\mathbb{Q}}.$$

What it is: Theorem ?? . π is transcendental by non-discharge.

22. **Half-turn witness**

$$R_{\pi}(z) = -z.$$

What it is: Definition 5.1. Exact antipodal action of the half-turn on the unit ORS orbit.

23. **Oriented boundary irreconcilability sign pattern**

$$m(z) \text{ changes sign across the wall.}$$

What it is: The sign-changing mass profile inherited from CF08 that realizes oriented boundary irreconcilability in 5D.

24. **Chirality as the 5D effective invariant**

$$5\text{D boundary half-turn} \equiv \text{chirality.}$$

What it is: Theorem 6.1. When maximal opposition is expressed as an oriented 5D boundary, the effective invariant is chirality.

25. Nielsen–Ninomiya closure obstruction

finite periodic closure $\Rightarrow$ paired chiral sectors.

What it is: Theorem 7.1. Under locality, periodicity, and finite algebraic closure, a single isolated chiral half-turn sector cannot appear without its partner.

26. Domain-wall evasion mechanism

domain wall $\Rightarrow$ asymptotic separation of the partner.

What it is: Theorem 8.1. The partner is not destroyed inside finite closure; it is displaced into the auxiliary-wall sector.

27. Compact ORS completion restatement

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta).$$

What it is: Compact restatement of the continuous ORS family in the final formalism summary.

28. Compact primitive half-turn restatement

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

What it is: Compact restatement of the primitive half-turn constant in the final formalism summary.

29. Compact opposite-pole attainment restatement

$$R_\pi = -I.$$

What it is: Compact restatement that the half-turn sends the invariant to its opposite pole.

30. Compact Euler identity restatement

$$e^{i\pi} = -1.$$

What it is: Compact restatement of half-turn closure on the complex carrier.

31. Compact transcendence restatement

$$\pi \notin \overline{\mathbb{Q}}^{\text{alg}}.$$

What it is: Compact restatement of the transcendence conclusion in the final formalism summary.

32. Compact maximal-irreconcilability identification

half-turn $\equiv$ maximal geometric irreconcilability.

What it is: Compact restatement of Theorem 5.1.

33. Compact chirality identification

5D boundary half-turn $\equiv$ chirality.

What it is: Compact restatement of the 5D effective-invariant identification.

34. Compact finite-closure obstruction statement

finite periodic closure $\Rightarrow$ paired chiral sectors.

What it is: Compact restatement of the Nielsen–Ninomiya obstruction in invariant language.

35. Compact domain-wall separation statement

domain wall $\Rightarrow$ asymptotic separation of the partner.

What it is: Compact restatement of the domain-wall evasion mechanism.

16 CF14 - Principle of Stationary Action as Temporal Effective Invariant

1. Primitive ORS quarter-turn seed

$$R(1, 0) = (0, 1), \quad R(0, 1) = (-1, 0).$$

What it is: The first discrete orthogonal articulation step inherited from the CF000 $\rightarrow$ CF00 spine.

2. Atomic orthogonal articulation operator

$$i^2 = -1.$$

What it is: Algebraic witness of the primitive quarter-turn already earned upstream and used here as the seed of temporal orthogonal articulation.

3. Least half-turn completion/opening constant

$$\pi.$$

What it is: The least completion/opening measure of continuous opposite-pole attainment inherited from CF00 and used here as the accumulated orthogonal-articulation measure.

4. Admissible path

$$\gamma : [t_i, t_f] \rightarrow \mathcal{Q}, \quad \gamma(t_i) = q_i, \quad \gamma(t_f) = q_f.$$

What it is: Definition of an admissible path between fixed endpoint configurations.

5. Endpoint-fixed admissible variation

$$\gamma_\varepsilon(t_i) = q_i, \quad \gamma_\varepsilon(t_f) = q_f, \quad \gamma_0 = \gamma.$$

What it is: Definition of an endpoint-fixed admissible variation family of paths.

6. Articulation-cost density

$$L_{\mathcal{I} \setminus \square}(q, \dot{q}, t).$$

What it is: Local temporal-domain cost density measuring the rate of accumulated invariant-bearing orthogonal articulation.

7. Total articulation cost / action

$$S_{\mathcal{I} \setminus \square}[\gamma] := \int_{t_i}^{t_f} L_{\mathcal{I} \setminus \square}(q, \dot{q}, t) dt.$$

What it is: Definition of total articulation cost along an admissible path.

8. Temporal effective-invariant rule

$$\delta S_{\mathcal{I} \setminus \square}[\gamma] = 0.$$

What it is: Primitive stationarity rule on endpoint-fixed admissible paths.

9. Primitive stationarity theorem

$$\delta S_{\mathcal{I} \setminus \square}[\gamma] = 0 \quad \text{for all endpoint-fixed admissible variations.}$$

What it is: Theorem ?? . A selected admissible path carries no avoidable first-order excess in accumulated articulation cost.

10. Endpoint-fixed variation ansatz

$$q_\varepsilon(t) = q(t) + \varepsilon \eta(t), \quad \eta(t_i) = \eta(t_f) = 0.$$

What it is: Standard variational ansatz used in the Euler–Lagrange derivation.

11. First variation of the action

$$\delta S_{\mathcal{I} \setminus \square} = \int_{t_i}^{t_f} \left(\frac{\partial L_{\mathcal{I} \setminus \square}}{\partial q} \eta + \frac{\partial L_{\mathcal{I} \setminus \square}}{\partial \dot{q}} \dot{\eta} \right) dt.$$

What it is: First-variation formula before integration by parts.

12. **Integrated-by-parts first variation**

$$\delta S_{\mathcal{I} \setminus \sqsubseteq} = \int_{t_i}^{t_f} \left(\frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial q} - \frac{d}{dt} \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial \dot{q}} \right) \eta dt.$$

What it is: First-variation formula after using endpoint-fixed boundary conditions.

13. **Euler–Lagrange equation**

$$\frac{d}{dt} \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial \dot{q}} - \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial q} = 0.$$

What it is: Theorem ?? . Local path-level non-discharge balance law.

14. **Propagation-resistance limb**

$$\frac{d}{dt} \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial \dot{q}}.$$

What it is: Path-level propagation-resistance term in the Euler–Lagrange balance.

15. **Configuration-burden limb**

$$\frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial q}.$$

What it is: Path-level configuration-burden term in the Euler–Lagrange balance.

16. **Fixed-energy conservative realization (Maupertuis)**

$$\delta \int p \cdot dq = 0 \quad \text{at fixed } E.$$

What it is: Fixed-energy conservative descendant of the primitive stationarity rule.

17. **Mechanical realization of the articulation-cost density**

$$L_{\mathcal{I} \setminus \sqsubseteq} \rightsquigarrow T - V.$$

What it is: Realization of the primitive cost density in the classical mechanical sector.

18. **Hamilton’s principle**

$$\delta \int_{t_i}^{t_f} (T - V) dt = 0.$$

What it is: Classical fixed-endpoint realization of the primitive stationarity rule.

19. **Thermodynamic internal-energy stationarity**

$$\delta U = 0 \quad \text{at fixed } (S, V, \dots).$$

What it is: Thermodynamic-potential realization on the appropriate constraint surface.

20. **Helmholtz free-energy stationarity**

$$\delta F = 0 \quad \text{at fixed } (T, V, \dots).$$

What it is: Thermodynamic-potential realization under fixed temperature and volume.

21. **Enthalpy stationarity**

$$\delta H = 0 \quad \text{at fixed } (S, P, \dots).$$

What it is: Thermodynamic-potential realization under fixed entropy and pressure.

22. **Gibbs free-energy stationarity**

$$\delta G = 0 \quad \text{at fixed } (T, P, \dots).$$

What it is: Thermodynamic-potential realization under fixed temperature and pressure.

23. **Onsager dissipation functional**

$$\Phi(J) := \frac{1}{2} \sum_{i,j} L_{ij} J_i J_j.$$

What it is: Dissipation functional for the irreversible near-equilibrium M-limb realization.

24. **Onsager reciprocity**

$$L_{ij} = L_{ji}.$$

What it is: Symmetry of the Onsager coefficients inherited from time-reversal symmetry of the M-limb.

25. **Entropy-production rate**

$$\dot{S} = \sum_i X_i J_i.$$

What it is: Irreversible entropy-production rate written in force-flux form.

26. **Onsager stationarity condition**

$$\frac{\partial}{\partial J_k} (\dot{S} - \Phi) = 0.$$

What it is: Variational selection law for irreversible fluxes at fixed thermodynamic forces.

27. **Onsager constitutive relations**

$$X_k = \sum_j L_{kj} J_j.$$

What it is: Linear irreversible constitutive law obtained from Onsager stationarity.

28. **Generic local field action**

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x) d^d x.$$

What it is: Generic field-theoretic realization of the primitive stationary-action rule.

29. **Field Euler–Lagrange equation**

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

What it is: Local field-space non-discharge balance law.

30. **Maxwell action**

$$S_{\text{Maxwell}}[A] = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} d^4 x.$$

What it is: Gauge-field realization of stationary action.

31. **Einstein–Hilbert action**

$$S_{\text{EH}}[g] = \frac{1}{16\pi G} \int R \sqrt{-g} d^4 x.$$

What it is: Gravitational realization of stationary action.

32. **Schwinger quantum action principle**

$$\delta \langle q_f, t_f \mid q_i, t_i \rangle = \frac{i}{\hbar} \langle q_f, t_f \mid \delta S \mid q_i, t_i \rangle.$$

What it is: Operator-level quantum descendant of the stationary-action law.

33. **Feynman path integral**

$$\langle q_f | e^{-iHt/\hbar} | q_i \rangle = \int \mathcal{D}[q] e^{\frac{i}{\hbar} S[q]}.$$

What it is: Quantum path-space realization of the stationary-action rule.

34. **Rayleigh–Ritz stationarity**

$$\delta \langle \psi | H | \psi \rangle = 0 \quad \text{subject to } \langle \psi | \psi \rangle = 1.$$

What it is: Variational quantum state-space descendant selecting stationary energy-expectation states.

35. **Time-dependent variational principle residual**

$$\delta \left\| \left(i\hbar \frac{d}{dt} - H \right) |\psi(\lambda(t))\rangle \right\| = 0.$$

What it is: State-manifold dynamical descendant of the quantum stationary-action law.

36. **Compact primitive stationarity restatement**

$$\delta S_{\mathcal{I} \setminus \underline{\square}} = 0.$$

What it is: Compact restatement of the root temporal effective-invariant rule.

37. **Compact Euler–Lagrange restatement**

$$\frac{d}{dt} \frac{\partial L_{\mathcal{I} \setminus \underline{\square}}}{\partial \dot{q}} - \frac{\partial L_{\mathcal{I} \setminus \underline{\square}}}{\partial q} = 0.$$

What it is: Compact restatement of the local path-level balance law.

38. **Compact Onsager realization restatement**

$$\frac{\partial}{\partial J_k} (\dot{S} - \Phi) = 0 \quad \implies \quad X_k = \sum_j L_{kj} J_j.$$

What it is: Compact restatement of the dissipative M-limb variational law.

39. **Compact field-action restatement**

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x) d^d x.$$

What it is: Compact restatement of generic local field-action stationarity.

40. **Compact Maxwell realization restatement**

$$S_{\text{Maxwell}}[A] = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} d^4x.$$

What it is: Compact restatement of the gauge-field descendant.

41. **Compact Einstein–Hilbert realization restatement**

$$S_{\text{EH}}[g] = \frac{1}{16\pi G} \int R \sqrt{-g} d^4x.$$

What it is: Compact restatement of the gravitational descendant.

42. **Compact Schwinger-principle restatement**

$$\delta \langle q_f, t_f \mid q_i, t_i \rangle = \frac{i}{\hbar} \langle q_f, t_f \mid \delta S \mid q_i, t_i \rangle.$$

What it is: Compact restatement of the operator-level quantum descendant.

43. **Compact path-integral restatement**

$$\langle q_f \mid e^{-iHt/\hbar} \mid q_i \rangle = \int \mathcal{D}[q] e^{\frac{i}{\hbar} S[q]}.$$

What it is: Compact restatement of the quantum path-space descendant.

17 CF15 - Noether's Theorem as Invariant Articulation Cost Under Symmetry

1. **CF14 selected-path balance law**

$$\frac{d}{dt} \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial \dot{q}} = \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial q}.$$

What it is: The inherited CF 14 two-pole path-balance law used as the starting point of CF15.

2. **Articulation-cost functional**

$$S_{\mathcal{I} \setminus \sqsubseteq}[\gamma] = \int_{t_i}^{t_f} L_{\mathcal{I} \setminus \sqsubseteq}(q, \dot{q}, t) dt.$$

What it is: Total articulation burden accumulated along a selected path.

3. **Propagation-resistance limb**

$$P := \frac{\partial L_{\mathcal{I} \setminus \sqsubseteq}}{\partial \dot{q}}.$$

What it is: The VDM-derived local burden carrier measuring resistance to changes in traversal rate.

4. **Propagation-resistance balance law**

$$\frac{dP}{dt} = \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial q}.$$

What it is: Path-balance law rewritten in terms of the propagation-resistance limb.

5. **Zero-cost direction condition**

$$\delta_\xi L_{\mathcal{I} \setminus \Xi} := \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial q} \cdot \xi + \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial \dot{q}} \cdot \dot{\xi} = \frac{d}{dt} \Lambda(q, t).$$

What it is: Definition ???. A smooth direction along which the first-order articulation cost is only a total time derivative.

6. **VDM-derived Noether charge**

$$Q_\xi := P \cdot \xi - \Lambda = \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial \dot{q}} \cdot \xi - \Lambda.$$

What it is: The VDM-derived burden carried in a zero-cost particle-sector direction.

7. **Field-sector articulation-cost functional**

$$S_{\mathcal{I} \setminus \Xi}[\phi] = \int \mathcal{L}_{\mathcal{I} \setminus \Xi}(\phi, \partial_\mu \phi, x) d^4x.$$

What it is: Field realization of the articulation-cost functional.

8. **Field-gradient burden density**

$$\Pi^\mu := \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial(\partial_\mu \phi)}.$$

What it is: The field-sector analogue of the propagation-resistance limb.

9. **Zero-cost field-variation condition**

$$\delta \mathcal{L}_{\mathcal{I} \setminus \Xi} = \partial_\mu K^\mu.$$

What it is: Definition of a field variation that costs only an admissible boundary flux.

10. **VDM-derived Noether current**

$$j^\mu := \Pi^\mu \delta \phi - K^\mu = \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial(\partial_\mu \phi)} \delta \phi - K^\mu.$$

What it is: The VDM-derived local burden density in a zero-cost field direction.

11. **One-parameter symmetry-group condition**

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_{\mathcal{I} \setminus \Xi}(q_\epsilon, \dot{q}_\epsilon, t) = \frac{d}{dt} \Lambda(q, t).$$

What it is: Definition ?? . Infinitesimal condition for a one-parameter zero-cost family.

12. **Infinitesimal symmetry generator**

$$\xi(q, t) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} q_\epsilon.$$

What it is: Generator of the one-parameter zero-cost family.

13. **Noether conservation law — particle sector**

$$\frac{d}{dt} Q_\xi = 0.$$

What it is: Theorem ?? . Conservation of the burden carried in a zero-cost particle-sector direction.

14. **Intermediate derivative of the particle-sector charge**

$$\frac{dQ_\xi}{dt} = \frac{d}{dt} \left(\frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial \dot{q}} \right) \cdot \xi + \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial \dot{q}} \cdot \dot{\xi} - \dot{\Lambda}.$$

What it is: Expansion step used in the proof of the particle-sector theorem.

15. **Particle-sector conservation step from the zero-cost condition**

$$\frac{dQ_\xi}{dt} = \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial q} \cdot \xi + \frac{\partial L_{\mathcal{I} \setminus \Xi}}{\partial \dot{q}} \cdot \dot{\xi} - \dot{\Lambda} = \delta_\xi L_{\mathcal{I} \setminus \Xi} - \dot{\Lambda} = 0.$$

What it is: Final proof chain showing exact conservation.

16. **Field-sector balance law**

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial \phi} = 0.$$

What it is: Field analogue of the CF 14 stationarity closure, derived by applying the same stationarity rule to the field functional.

17. **Field-sector variational identity**

$$\delta S_{\mathcal{I} \setminus \Xi} = \int \left[\frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \right) \right] \delta \phi d^4 x = 0.$$

What it is: Variational identity from which the field balance law follows.

18. **Noether current conservation — field sector**

$$\partial_\mu j^\mu = 0.$$

What it is: Theorem ?? . Local conservation of the burden current in a zero-cost field direction.

19. **Field-sector divergence expansion**

$$\partial_\mu j^\mu = \partial_\mu \left(\frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \right) \delta \phi + \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \partial_\mu (\delta \phi) - \partial_\mu K^\mu.$$

What it is: Product-rule expansion used in the field-sector proof.

20. **Field zero-cost identity**

$$\delta \mathcal{L}_{\mathcal{I} \setminus \Xi} = \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \partial_\mu (\delta \phi) = \partial_\mu K^\mu.$$

What it is: Field zero-cost condition substituted into the divergence proof.

21. **Field-sector conservation step on shell**

$$\partial_\mu j^\mu = \left[\partial_\mu \left(\frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial \phi} \right] \delta \phi = 0.$$

What it is: Final on-shell conservation step for the field current.

22. **General metriplectic equation of motion**

$$\frac{dF}{dt} = \{F, H\}_J + [F, \Sigma]_M.$$

What it is: Inherited CF 01 equation of motion for any admissible observable.

23. **J-limb exact conservation law**

$$\frac{d}{dt} Q_\xi = \{Q_\xi, H\}_J = 0.$$

What it is: Theorem ?? . Exact conservation of Hamiltonian Noether charges on a pure J-limb symplectic leaf.

24. **Hamiltonian-layer zero-cost condition**

$$\{Q_\xi, H\}_J = 0.$$

What it is: Hamiltonian-layer expression of the zero-cost condition for a symmetry generator.

25. **Pure J-limb frozen-entropy condition**

$$[Q_\xi, \Sigma]_M = 0.$$

What it is: Vanishing of the M-limb term under pure reversible evolution.

26. **General metriplectic Noether-charge evolution**

$$\frac{d}{dt}Q_\xi = \{Q_\xi, H\}_J + [Q_\xi, \Sigma]_M.$$

What it is: Theorem ?? . Full drift law for a Hamiltonian Noether charge in a metriplectic system.

27. **M-limb symmetry-breaking law**

$$\frac{d}{dt}Q_\xi = [Q_\xi, \Sigma]_M.$$

What it is: Theorem ?? . Drift law when Q_ξ is a symmetry of H .

28. **Entropy-production law**

$$\frac{d\Sigma}{dt} = [\Sigma, \Sigma]_M \geq 0.$$

What it is: Inherited CF 02 entropy law, explicitly linked here to M-limb symmetry breaking.

29. **Aligned M-limb breaking condition**

$$[Q_\xi, \Sigma]_M = \kappa_\xi(x) [\Sigma, \Sigma]_M.$$

What it is: Corollary ?? hypothesis for aligned charge drift.

30. **Aligned-drift law**

$$\frac{d}{dt}Q_\xi = \kappa_\xi(x) \frac{d\Sigma}{dt}.$$

What it is: Corollary ?? . Charge drift proportional to the entropy-production rate.

31. **Normalized aligned case**

$$\kappa_\xi = 1.$$

What it is: Special aligned case where the free Noether charge drifts exactly at the entropy-production rate.

32. **Coordinate form of the M-bracket**

$$[F, G]_M = M^{\mu\nu} \partial_\mu F \partial_\nu G.$$

What it is: Coordinate representation of the M-bracket used for the irreversible-vector-field construction.

33. Irreversible entropy-producing vector field

$$v_M^\mu := M^{\mu\nu} \partial_\nu \Sigma.$$

What it is: Definition ???. Irreversible vector field generated by the entropy gradient through the M-operator.

34. Common Noether kernel

$$\mathcal{K}_x := \bigcap_{Q_\xi \in \mathcal{Q}_N^{\text{free}}} \ker(dQ_\xi|_x) \subset T_x \mathcal{M}.$$

What it is: Definition ???. Common kernel of all free Noether-charge differentials.

35. Equilibrium condition

$$[\Sigma, \Sigma]_M = 0.$$

What it is: Equilibrium condition inherited from CF 02 and used to prove restoration of exact Hamiltonian symmetry content.

36. Coordinate entropy-quadratic form

$$[\Sigma, \Sigma]_M = M^{\mu\nu} \partial_\mu \Sigma \partial_\nu \Sigma.$$

What it is: Coordinate form used to show that zero entropy production forces vanishing irreversible vector field.

37. Equilibrium irreversible-inertia condition

$$M^{\mu\nu} \partial_\nu \Sigma = 0.$$

What it is: Derived consequence of equilibrium for a symmetric positive semidefinite M-operator.

38. Vanishing irreversible vector field at equilibrium

$$v_M^\mu = 0.$$

What it is: Equilibrium consequence showing the irreversible sector is inert.

39. M-bracket as pairing with the irreversible vector field

$$[F, \Sigma]_M = dF(v_M).$$

What it is: Coordinate interpretation used in the transversality proof.

40. Equilibrium symmetry maximization law

$$\frac{dQ_\xi}{dt} = [Q_\xi, \Sigma]_M = 0.$$

What it is: Theorem ???. At equilibrium every Hamiltonian Noether charge is exactly conserved.

41. **Transversality equivalence**

$$[Q_\xi, \Sigma]_M(x) = 0 \text{ for all free Noether charges} \iff v_M(x) \in \mathcal{K}_x.$$

What it is: Lemma ?? . Characterization of simultaneous vanishing M-drift for all free Noether charges.

42. **Active non-equilibrium condition on an open set**

$$[\Sigma, \Sigma]_M(x) > 0 \quad \text{for all } x \in B.$$

What it is: Hypothesis of Theorem ?? defining active unresolved M-limb burden on an open set.

43. **Boundary-hosted non-equilibrium symmetry reduction**

$$\exists Q_\xi \in \mathcal{Q}_N^{\text{free}} \text{ such that } [Q_\xi, \Sigma]_M(x) \neq 0 \text{ for some } x \in B.$$

What it is: Theorem ?? . In a closed realized sector with active non-equilibrium, at least one free Noether charge must drift.

44. **Energy charge**

$$Q_{\partial_t} = \frac{\partial L_{\mathcal{T} \setminus \Xi}}{\partial \dot{q}} \cdot \dot{q} - L_{\mathcal{T} \setminus \Xi}.$$

What it is: Realized descendant for time-translation zero-cost structure; recognized in standard physics as the Hamiltonian/energy.

45. **Momentum charge**

$$Q_{\hat{n}} = \frac{\partial L_{\mathcal{T} \setminus \Xi}}{\partial \dot{q}} \cdot \hat{n}.$$

What it is: Realized descendant for uniform carrier translation; recognized as linear momentum in direction $\hat{n}$.

46. **Angular-momentum charge**

$$Q_{\hat{n}} = \frac{\partial L_{\mathcal{T} \setminus \Xi}}{\partial \dot{q}} \cdot (\hat{n} \times q).$$

What it is: Realized descendant for rigid carrier rotation about axis $\hat{n}$; recognized as angular momentum.

47. **Electric-current realization from ORS phase redundancy**

$$j_{\text{EM}}^\mu = \frac{\partial \mathcal{L}_{\mathcal{T} \setminus \Xi}}{\partial (\partial_\mu \phi)}(i\phi) + \text{h.c.}$$

What it is: Realized descendant for the $U(1)$ gauge-sector zero-cost direction; recognized as the electric current.

48. **Non-abelian zero-cost fibre variation**

$$\delta\phi = iT^a\phi.$$

What it is: Infinitesimal traceless fibre-rotation direction for the realized N -mode degenerate fibre sector.

49. **Non-abelian colour current**

$$j^{\mu,a} = \frac{\partial \mathcal{L}_{\mathcal{I} \setminus \Xi}}{\partial (\partial_\mu \phi)} (iT^a \phi) + \text{h.c.}$$

What it is: Realized descendant for non-abelian fibre degeneracy; recognized as the colour current.

50. **Stress-energy tensor**

$$T^{\mu\nu} := \frac{-2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L})}{\delta g_{\mu\nu}}.$$

What it is: Realized descendant of diffeomorphism zero-cost structure; recognized as the stress-energy tensor.

51. **Covariant stress-energy conservation**

$$\nabla_\mu T^{\mu\nu} = 0.$$

What it is: Standard-physics recognition of the geometric Noether conservation law.

52. **Inverse-Noether condition**

$$\{H, Q\}_J = 0.$$

What it is: Proposition ?? . Condition for a conserved observable to generate a one-parameter symmetry flow on a nondegenerate symplectic leaf.

53. **Compact zero-cost conservation mechanism**

$$\text{zero-cost direction} \implies \text{fixed burden.}$$

What it is: Conceptual closure statement of CF15: the invariant cannot create burden in a direction it does not pay for and cannot remove it without discharge.

18 CF16 - Electroweak Fiber Splitting, Chiral Weak Coupling, Neutral Gauge Mixing, and Broken Gauge–Scalar Structure

1. Minimal electroweak-hosting fiber

$$\mathcal{F}_{\text{EW}} \cong U(1)_{\text{comp}} \times SU(2)_L.$$

What it is: Theorem ??, clause 1. The minimal realized electroweak-hosting fiber.

2. Minimal electroweak order doublet

$$\Phi = \begin{pmatrix} \phi_+ \\ \phi_0 \end{pmatrix}.$$

What it is: Definition ?. The minimal realized electroweak order parameter.

3. Weak-isospin action on the charged component

$$T_3 \phi_+ = +\frac{1}{2} \phi_+.$$

What it is: Definition ?. The charged component carries $+\frac{1}{2}$ weak-isospin.

4. Weak-isospin action on the neutral component

$$T_3 \phi_0 = -\frac{1}{2} \phi_0.$$

What it is: Definition ?. The condensed neutral component carries $-\frac{1}{2}$ weak-isospin.

5. Neutrality condition for the condensed component

$$Q_{\text{em}} \phi_0 = 0.$$

What it is: Definition ?. The surviving residual neutral direction must annihilate the condensate.

6. Most general surviving neutral generator

$$Q = T_3 + cY.$$

What it is: Intermediate neutral-sector ansatz used in the proof of the residual electromagnetic charge law.

7. Companion-burden eigenvalue on the order doublet

$$Y \phi_+ = y_{\Phi} \phi_+, \quad Y \phi_0 = y_{\Phi} \phi_0.$$

What it is: The common companion-burden eigenvalue carried by the electroweak order doublet.

8. **Neutrality equation for the condensate**

$$0 = Q\phi_0 = \left(-\frac{1}{2} + cy_\Phi\right)\phi_0.$$

What it is: The condition fixing the coefficient c in the surviving neutral generator.

9. **Coefficient fixed by condensate neutrality**

$$c = \frac{1}{2y_\Phi}.$$

What it is: The coefficient forced by the neutrality condition.

10. **Normalized companion generator**

$$\tilde{Y} := \frac{Y}{y_\Phi}.$$

What it is: Intermediate normalization step used to fix the companion-generator convention.

11. **Residual electromagnetic charge law**

$$Q_{\text{em}} := T_3 + \frac{Y}{2}.$$

What it is: Definition ?? and Theorem ??, clause 3. The unique residual neutral charge operator.

12. **Single-condensate neutral covector**

$$n = (a, -b).$$

What it is: The single neutral covector seen by the condensate in the basis (W^3, B) .

13. **Quadratic neutral burden from the seen direction**

$$\nu^2(n \cdot x)^2 = \nu^2 x^\top (n n^\top) x, \quad x = (W^3, B)^\top.$$

What it is: Rank-one neutral broken-sector burden induced by a single condensate.

14. **Neutral broken-sector burden matrix**

$$\mathcal{M}_N^2 = \nu^2 \begin{pmatrix} a^2 & -ab \\ -ab & b^2 \end{pmatrix}.$$

What it is: Theorem ??, clause 4. Rank-one positive-semidefinite neutral mass/burden matrix.

15. **Electroweak order amplitude**

$$\nu := |\Phi|.$$

What it is: The condensate amplitude entering the neutral and charged mass formulas.

16. **Kernel direction of the neutral burden matrix**

$$(b, a).$$

What it is: The vector spanning the massless neutral direction of the rank-one neutral matrix.

17. **Residual massless neutral direction**

$$A = \sin \theta_{\text{EW}} W^3 + \cos \theta_{\text{EW}} B.$$

What it is: Theorem ??, clause 5. The surviving massless residual neutral direction.

18. **Orthogonal massive neutral direction**

$$Z = \cos \theta_{\text{EW}} W^3 - \sin \theta_{\text{EW}} B.$$

What it is: Theorem ??, clause 5. The orthogonal massive neutral direction.

19. **Electroweak mixing-angle cosine**

$$\cos \theta_{\text{EW}} = \frac{a}{\sqrt{a^2 + b^2}}.$$

What it is: Theorem ??, clause 5. The diagonalizing neutral-sector angle.

20. **Electroweak mixing-angle sine**

$$\sin \theta_{\text{EW}} = \frac{b}{\sqrt{a^2 + b^2}}.$$

What it is: Theorem ??, clause 5. The diagonalizing neutral-sector angle.

21. **Charged weak-sector mass law**

$$m_W^2 = \nu^2 a^2.$$

What it is: Theorem ??, clause 6. The common charged weak mass generated by the electroweak order parameter.

22. **Massive neutral-sector mass law**

$$m_Z^2 = \nu^2 (a^2 + b^2).$$

What it is: Theorem ??, clause 6. The massive neutral-sector mass.

23. Charged/neutral mass relation

$$m_W = m_Z \cos \theta_{EW}.$$

What it is: Theorem ??, clause 6. The structural/tree-level W/Z mass law.

24. Weak-host coefficient identity for the charged sector

$$a^2.$$

What it is: The common weak-host quadratic burden coefficient assigned to the pure weak directions before neutral mixing.

25. Residual electromagnetic charge law restated

$$Q_{em} = T_3 + \frac{Y}{2}.$$

What it is: The standard electroweak recognition recovered as a downstream reading of the VDM closure.

26. Neutral-sector diagonalization pair

$$(W^3, B) \longrightarrow (A, Z).$$

What it is: The structural neutral-sector splitting induced by the condensate.

27. Broken gauge–scalar split

tangential broken directions $\oplus$ normal mode of Φ .

What it is: Theorem ??, clause 7. The tangential broken directions are gauge-carried, while the normal mode remains the physical scalar.

28. Compact electroweak closure restatement

$$\mathcal{F}_{EW} \cong U(1)_{\text{comp}} \times SU(2)_L.$$

What it is: Compact restatement of the minimal electroweak-hosting fiber in the conclusions.

29. Compact electromagnetic charge restatement

$$Q_{em} = T_3 + \frac{Y}{2}.$$

What it is: Compact restatement of the residual electromagnetic charge law in the conclusions.

30. **Compact structural Weinberg-angle restatement**

$$\theta_{\text{EW}}.$$

What it is: The neutral-sector diagonalizing angle defined structurally by the rank-one broken neutral decomposition.

31. **Compact residual electromagnetism restatement**

$$A = \sin \theta_{\text{EW}} W^3 + \cos \theta_{\text{EW}} B.$$

What it is: Compact restatement of the residual massless neutral direction.

32. **Compact charged/neutral mass-law restatement**

$$m_W = m_Z \cos \theta_{\text{EW}}.$$

What it is: Compact restatement of the structural/tree-level W/Z mass relation.

19 CF17 - Non-Abelian Saturation, Flux-Tube Formation, Quantitative Confinement, String Breaking, and Deconfinement

1. **Confinement observable**

$$F_{\text{conf}}(R) := E_{\text{sep}}(R) - E_{\text{sep}}(0).$$

What it is: Definition ???. Excess articulation cost of a separated color-source configuration relative to the coincident neutralized configuration.

2. **Confining large-separation law**

$$F_{\text{conf}}(R) = \sigma_{\text{conf}} R + o(R).$$

What it is: Definition ???. A sector is confining if the excess cost grows linearly with positive slope at large separation.

3. **Positive confinement coefficient**

$$\sigma_{\text{conf}} > 0.$$

What it is: Positive string-tension condition in the confining regime.

4. **Tube-support functional**

$$\mathcal{E}_{\mathcal{T}}[\rho; R] = R \epsilon_{\perp}[\rho] + E_{\partial}[\rho] + E_{\text{mix}}[\rho].$$

What it is: Definition ???. Effective articulation-cost functional for a candidate flux-tube support profile.

5. **Main confinement scaling law**

$$F_{\text{conf}}(R) = \sigma_{\text{conf}} R + o(R), \quad \sigma_{\text{conf}} > 0.$$

What it is: Theorem ??, clause 4. Large-separation confinement law in the confined regime.

6. **Quantitative confinement law**

$$F_{\text{conf}}(R) = \sigma_{\text{conf}} R + O(1), \quad \sigma_{\text{conf}} := \epsilon_{\perp}[\rho_*] > 0.$$

What it is: Theorem ??. Linear large- R growth of the confinement observable for a minimizing tube profile ρ_* .

7. **Minimizing tube-profile substitution**

$$\mathcal{E}_{\mathcal{T}}[\rho_*; R] = R \epsilon_{\perp}[\rho_*] + E_{\partial}[\rho_*] + E_{\text{mix}}[\rho_*].$$

What it is: The minimizing branch of the tube-support functional used in the quantitative-confinement proof.

8. **Preferred-radius criterion**

$\epsilon_{\perp}(a)$ has an interior minimum with positive curvature.

What it is: Theorem ??. Condition for a preferred confinement radius when the transverse support family is parametrized by a radius a .

9. **String-breaking threshold inequality**

$$\sigma_{\text{conf}} R + O(1) \geq E_{\text{pair}}.$$

What it is: Corollary ??. Threshold at which the single-tube branch ceases to be the minimum admissible continuation.

10. **Strong-force effective stack**

$$(\mathcal{I} \setminus \sqsubseteq_{-1}, \mathcal{I} \setminus \sqsubseteq_{\text{host}}, \mathcal{I} \setminus \sqsubseteq_{\text{NA}}, \mathcal{I} \setminus \sqsubseteq_{\text{tube}}, \mathcal{I} \setminus \sqsubseteq_{\text{break}}).$$

What it is: Effective-stack chain for the confinement branch.

11. **Primitive invariant layer**

$$\mathcal{I} \setminus \sqsubseteq_{-1}.$$

What it is: Primitive non-discharge and saturation law.

12. **Gauge-host layer**

$$\mathcal{I} \setminus \sqsubseteq_{\text{host}}.$$

What it is: Admissible gauge/matter host layer.

13. **Non-abelian color-bearing layer**

$$\mathcal{I} \setminus \sqsubseteq_{\text{NA}}.$$

What it is: Already-earned non-abelian color sector.

14. **Confined filamentary-support layer**

$$\mathcal{I} \setminus \sqsubseteq_{\text{tube}}.$$

What it is: Confined flux-tube realization layer.

15. **Break / re-articulation layer**

$$\mathcal{I} \setminus \sqsubseteq_{\text{break}}.$$

What it is: Layer where the system re-articulates into new paired endpoints once tube growth ceases to be minimal.

16. **Asymptotic exclusion statement**

isolated free bulk color states are not admissible terminal realizations.

What it is: Theorem ?? . Free isolated color cannot terminate unresolved burden in a closed non-abelian sector.

17. **Flux concentration statement**

separated non-abelian color sources force concentration into a tube-supporting geometry.

What it is: Theorem ?? . Source separation forces a filamentary boundary-hosted support branch.

18. **Tube concentration law**

one-dimensional continuation along the separation axis with bounded transverse support.

What it is: Structural content of Lemma ?? . Minimal admissible continuation after diffuse bulk support saturates.

19. **Preferred-tube support law**

burden concentrates into a single support tube rather than disjoint bulk fragments.

What it is: Consequence of Theorem ?? .

20. **Tube-support local-minimum condition**

ρ_* is a local minimum of the effective support functional.

What it is: Condition defining the confined branch and used in the deconfinement criterion.

21. **Deconfinement criterion**

if ρ_* ceases to be a local minimum, the sector is deconfined.

What it is: Corollary ?? . Deconfinement is loss of the tube-supporting local minimum.

22. **Compact confinement closure restatement**

$$F_{\text{conf}}(R) = \sigma_{\text{conf}} R + o(R), \quad \sigma_{\text{conf}} > 0.$$

What it is: Compact restatement of the confinement law recognized downstream as color confinement with string tension.

23. **Compact string-tension definition**

$$\sigma_{\text{conf}} := \epsilon_{\perp}[\rho_*].$$

What it is: Compact restatement of the confinement slope as the per-length cost density of the minimizing tube profile.

24. **Compact string-breaking law**

$$\sigma_{\text{conf}} R + O(1) \geq E_{\text{pair}}.$$

What it is: Compact restatement of the threshold competition between tube growth and pair re-articulation.

25. **Compact deconfinement law**

deconfinement $\iff$ loss of the tube-supporting local minimum.

What it is: Compact restatement of the deconfinement criterion.

20 CF18 - Scale Re-Articulation, Effective Action, Hidden-Return Corrections, Running Couplings, Renormalization, and Corrected Symmetry Structure

1. **Reduced-description split**

$$q = (q_{\text{exp}}^{(\mu_{\text{RG}})}, q_{\text{hid}}^{(\mu_{\text{RG}})}).$$

What it is: Definition of the explicit/hidden split at a chosen scale μ_{RG} .

2. **Reduced-description projection**

$$\Pi_{\mu_{\text{RG}}} : q \mapsto q_{\text{exp}}^{(\mu_{\text{RG}})}.$$

What it is: The reduced-description map sending full degrees of freedom to the explicit sector at scale μ_{RG} .

3. Effective articulation-cost functional

$$S_{\text{eff}}[q_{\text{exp}}^{(\mu_{\text{RG}})}; \mu_{\text{RG}}] := S_{\text{exp}}[q_{\text{exp}}^{(\mu_{\text{RG}})}] + \Delta_{\text{hid}}[q_{\text{exp}}^{(\mu_{\text{RG}})}; \mu_{\text{RG}}].$$

What it is: Definition of the scale-indexed effective functional after hidden burden is re-expressed in visible form.

4. Hidden-return correction sum

$$\Delta_{\text{hid}} = \sum_{C \in \mathfrak{C}_{\mu_{\text{RG}}}} \Delta_C.$$

What it is: The induced correction term written as the sum over admissible hidden-return channels at scale μ_{RG} .

5. Beta-function definition

$$\beta_i(\mu_{\text{RG}}) := \mu_{\text{RG}} \frac{dg_i}{d\mu_{\text{RG}}}.$$

What it is: Definition of the beta function for any scale-indexed visible coefficient $g_i(\mu_{\text{RG}})$.

6. Fixed-point condition

$$\beta_i(\mu_*) = 0.$$

What it is: Fixed-point condition for coefficient g_i at scale μ_* .

7. Renormalization condition

$$\mathcal{O}_a^{\text{eff}}(\mu_0) = \mathcal{O}_a^{\text{phys}}.$$

What it is: Physical normalization condition fixing reduced observables at a reference scale μ_0 .

8. Counterterm

$$\delta S_{\text{ct}}[q_{\text{exp}}^{(\mu_{\text{RG}})}; \mu_{\text{RG}}].$$

What it is: Compensating visible term added so the reduced description satisfies the chosen normalization conditions.

9. Corrected-anomaly definition

$$\delta_{\xi}(S_{\text{eff}} + \delta S_{\text{ct}}) = \int \mathcal{A}_{\xi} d^d x, \quad \mathcal{A}_{\xi} \not\equiv 0.$$

What it is: Definition of a genuine surviving anomaly in the corrected effective layer.

10. Scale-indexed effective stack

$$(S_{\text{bare}}, \Pi_{\mu_{\text{RG}}}, S_{\text{eff}}(\mu_{\text{RG}}), g_i(\mu_{\text{RG}}), \delta S_{\text{ct}}(\mu_{\text{RG}}), \mathcal{J}_{\text{corr}}(\mu_{\text{RG}})).$$

What it is: Ordered effective stack for the radiative/scale branch.

11. **Effective functional closure law**

$$S_{\text{eff}} = S_{\text{exp}} + \Delta_{\text{hid}}.$$

What it is: Core reduced-law identity appearing in the main governing theorem.

12. **Running equation**

$$\mu_{\text{RG}} \frac{dg_i}{d\mu_{\text{RG}}} = \beta_i(\mu_{\text{RG}}).$$

What it is: Theorem ?? . Running law for any visible coefficient in the reduced effective functional.

13. **Fixed-point equivalence**

$$\beta_i = 0 \iff g_i(\mu_{\text{RG}}) \text{ is unchanged under the scale shift at the chosen effective layer.}$$

What it is: Fixed-point characterization from the running-coupling theorem.

14. **Renormalized effective functional**

$$S_{\text{eff}}^{\text{ren}} := S_{\text{eff}} + \delta S_{\text{ct}}.$$

What it is: Theorem ?? . Corrected reduced functional after compensating counterterms are added.

15. **Corrected symmetry preservation case**

$$\delta_\xi S_{\text{eff}}^{\text{ren}} = 0.$$

What it is: Theorem ?? , case (i). Exact symmetry survives in the corrected effective layer.

16. **Corrected drift / soft-breaking case**

$$\delta_\xi S_{\text{eff}}^{\text{ren}} \neq 0.$$

What it is: Theorem ?? , case (ii). The corrected variation is nonzero but removable within the admissible reduced layer.

17. **Anomalous case**

$$\delta_\xi S_{\text{eff}}^{\text{ren}} \text{ contains a surviving obstruction } \mathcal{A}_\xi.$$

What it is: Theorem ?? , case (iii). Genuine anomaly in the corrected effective layer.

18. **Compact full articulation law**

$$S_{\text{bare}}[q].$$

What it is: Bare/full articulation-cost functional at the unreduced level.

19. **Compact effective-action restatement**

$$S_{\text{eff}} = S_{\text{exp}} + \Delta_{\text{hid}}.$$

What it is: Compact restatement of the effective reduced law.

20. **Compact hidden-return correction restatement**

$$\Delta_{\text{hid}} = \sum_{C \in \mathfrak{C}_{\mu_{\text{RG}}}} \Delta_C.$$

What it is: Compact restatement of hidden-return re-expression.

21. **Compact beta-function restatement**

$$\beta_i(\mu_{\text{RG}}) := \mu_{\text{RG}} \frac{dg_i}{d\mu_{\text{RG}}}.$$

What it is: Compact restatement of running coefficients.

22. **Compact renormalized effective law**

$$S_{\text{eff}}^{\text{ren}} = S_{\text{eff}} + \delta S_{\text{ct}}.$$

What it is: Compact restatement of renormalized effective action.

23. **Compact anomaly restatement**

$$\delta_\xi(S_{\text{eff}} + \delta S_{\text{ct}}) = \int \mathcal{A}_\xi d^d x.$$

What it is: Compact restatement of corrected anomaly structure.